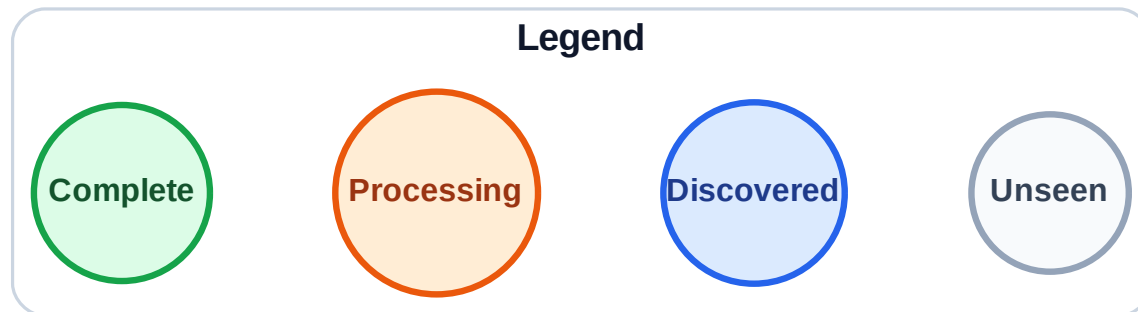
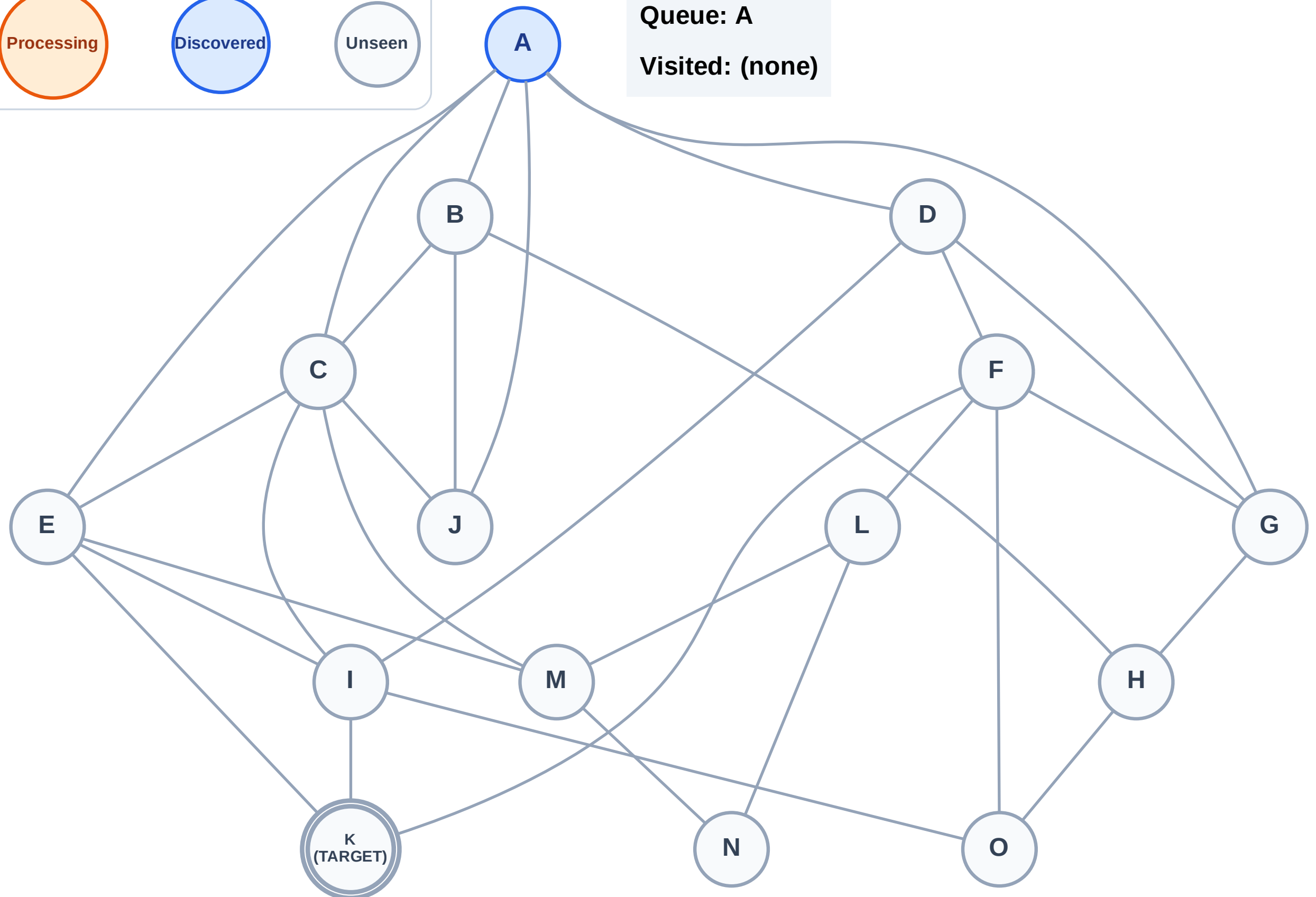


BFS Traversal

Step 1: Start at A (Seeking target node K)

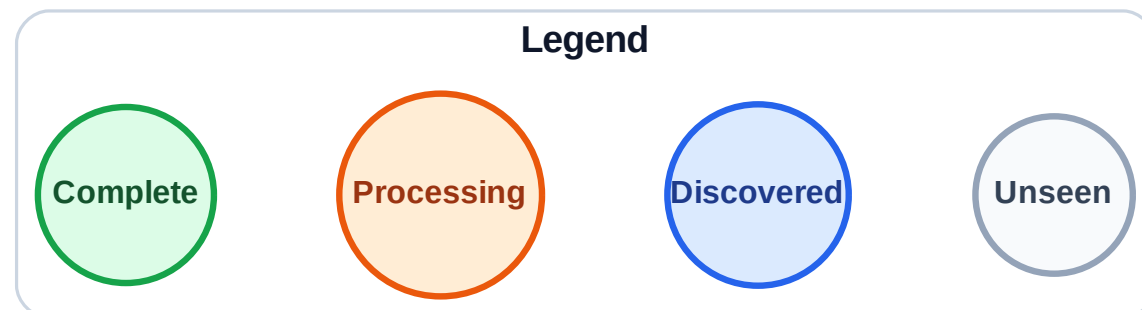


Queue: A
Visited: (none)

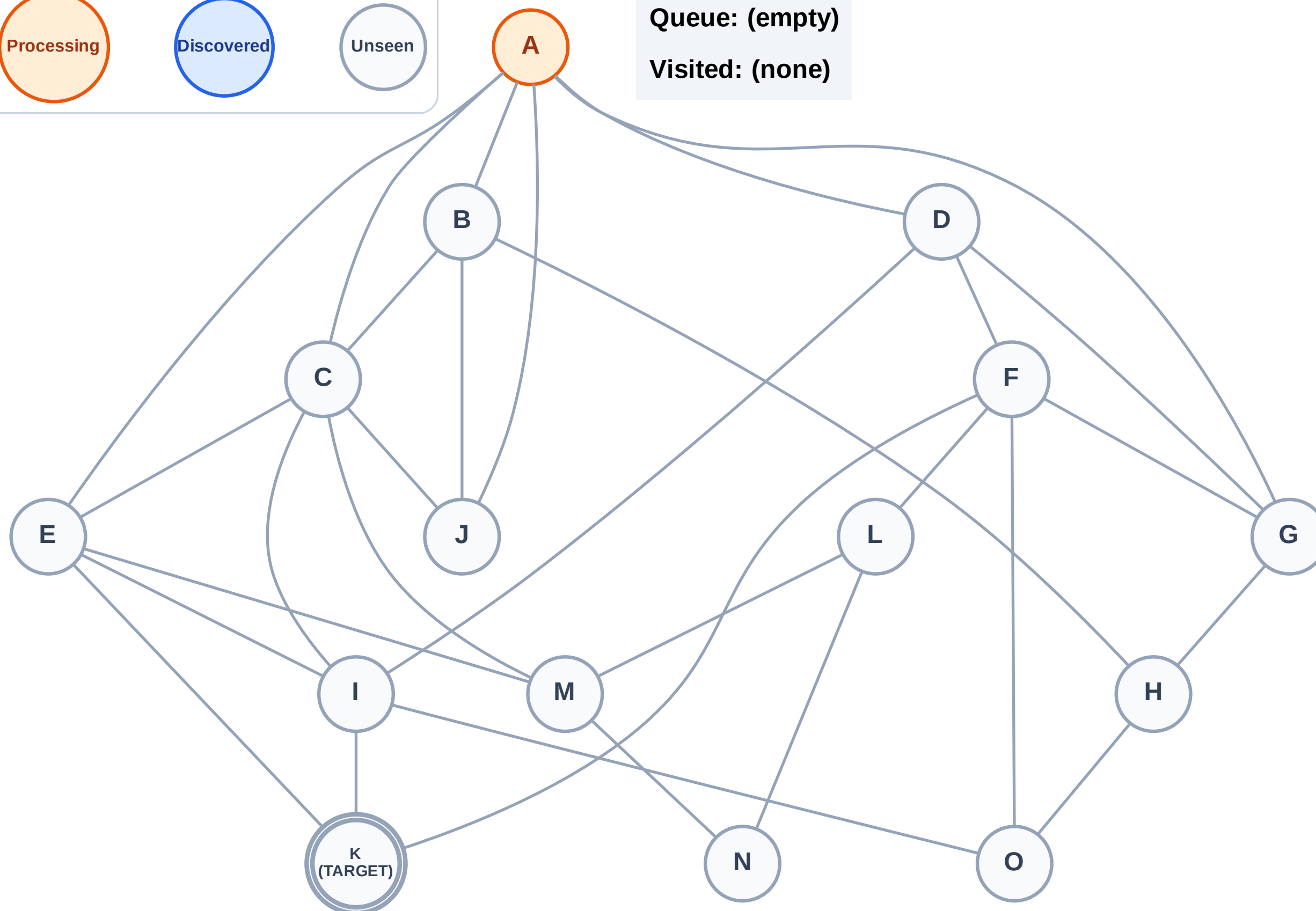


BFS Traversal

Step 2: Process node A



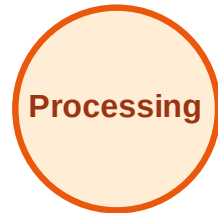
Queue: (empty)
Visited: (none)



BFS Traversal

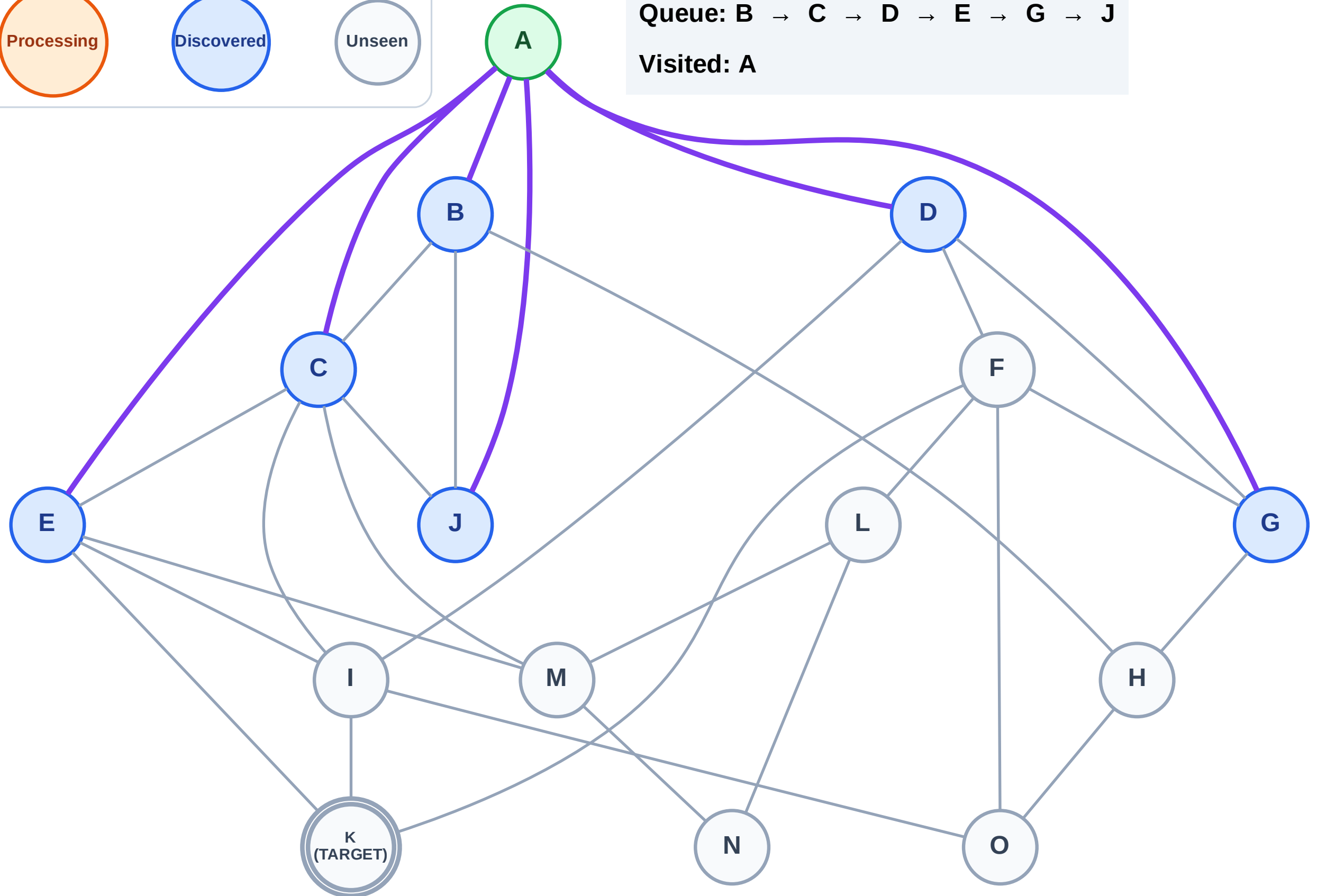
Step 3: Discover B, C, D, E, G, J from A

Legend



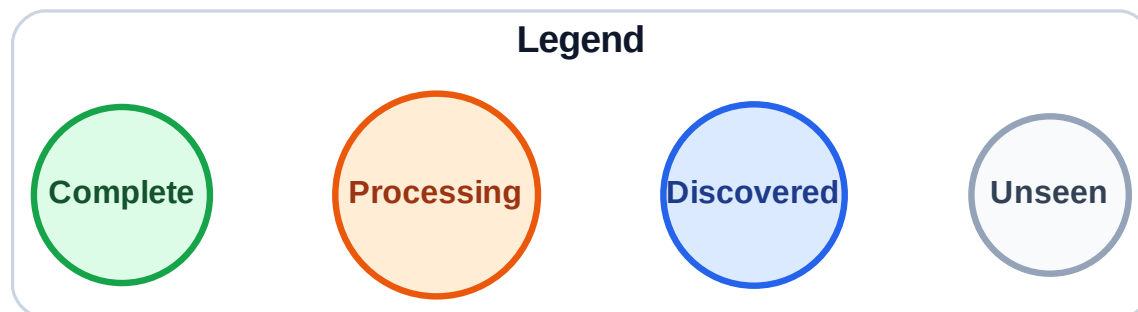
Queue: B → C → D → E → G → J

Visited: A

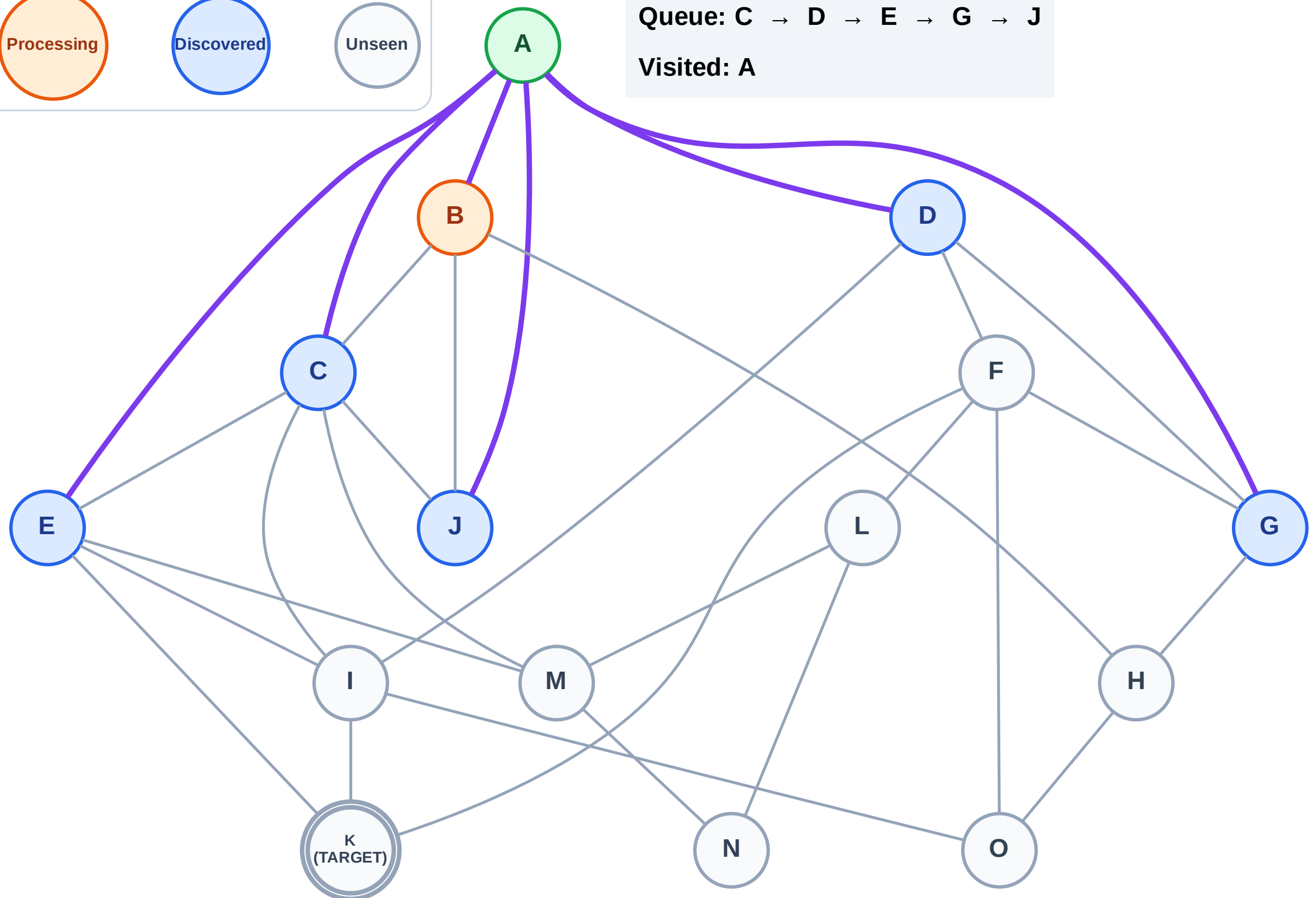


BFS Traversal

Step 4: Process node B



Queue: C → D → E → G → J
Visited: A



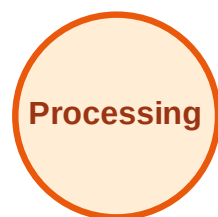
BFS Traversal

Step 5: Discover H from B

Legend



Complete



Processing



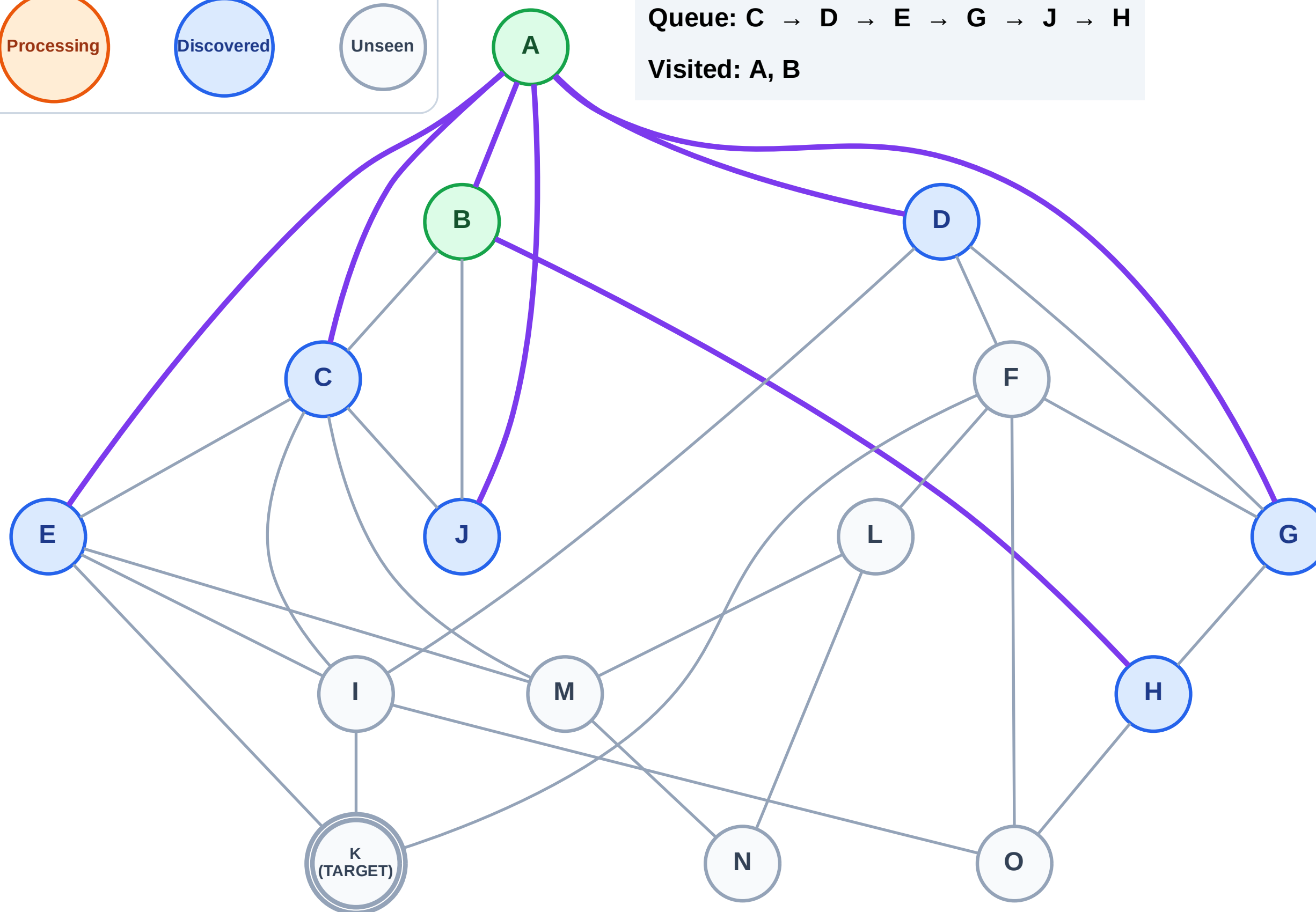
Discovered



Unseen

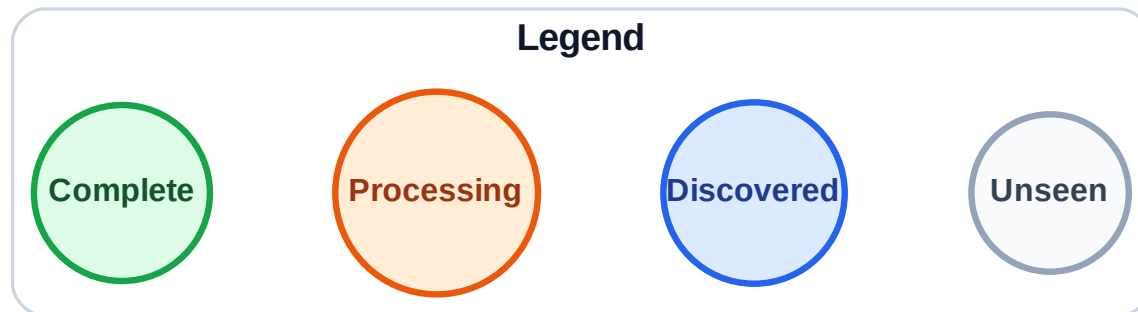
Queue: C → D → E → G → J → H

Visited: A, B



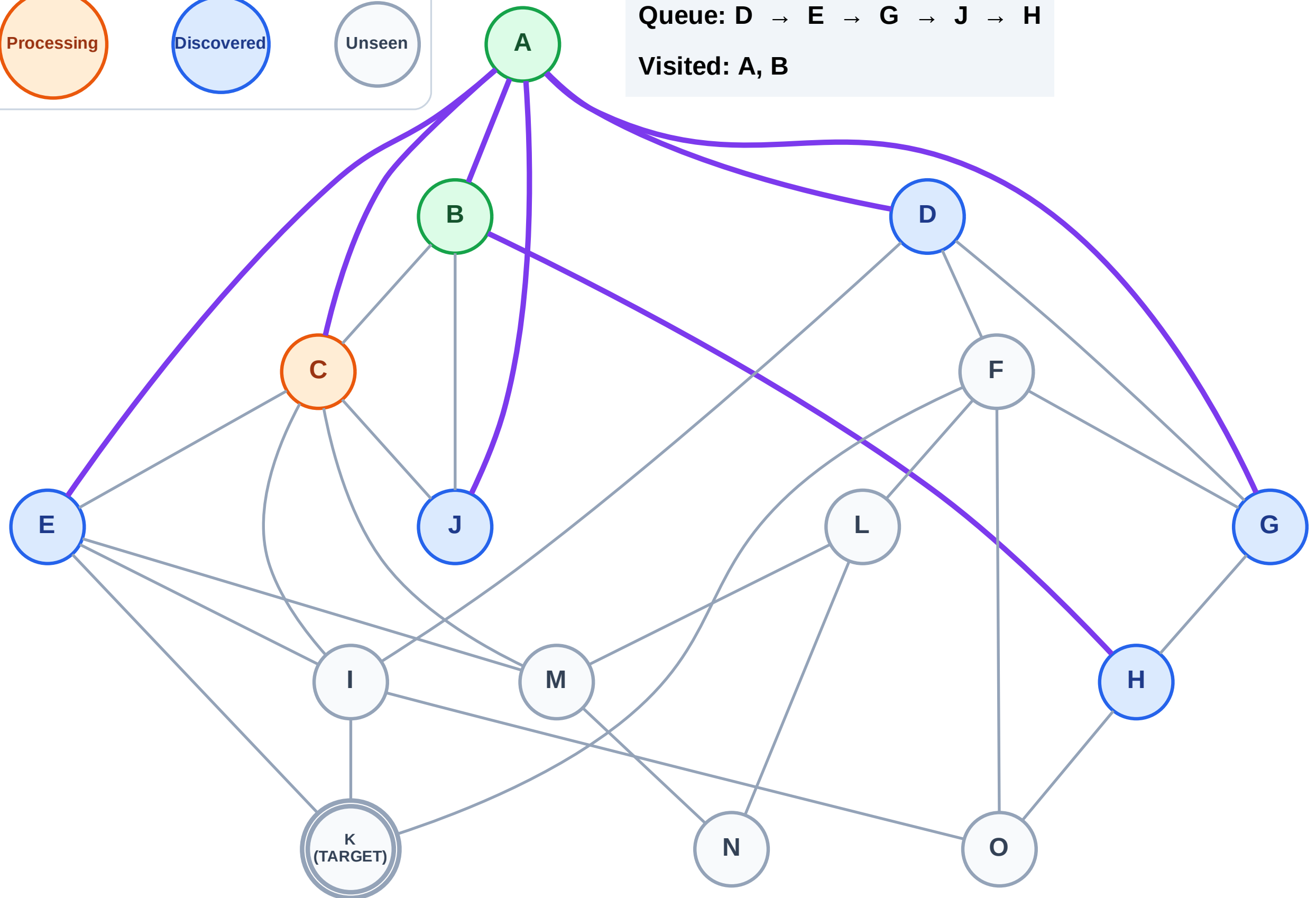
BFS Traversal

Step 6: Process node C



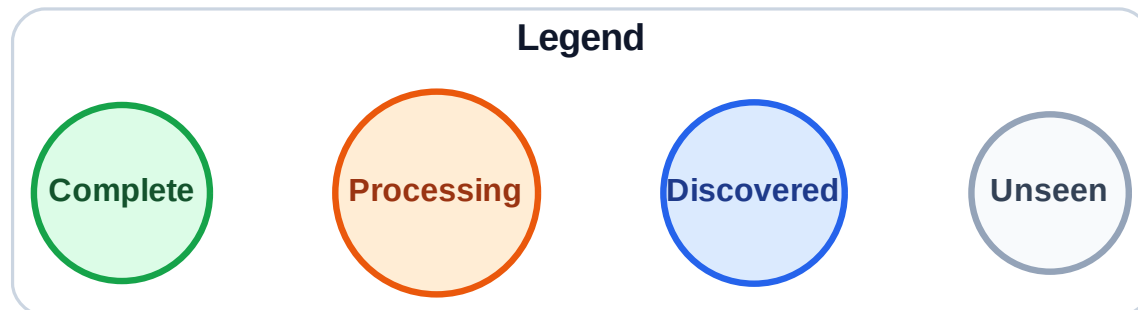
Queue: D → E → G → J → H

Visited: A, B



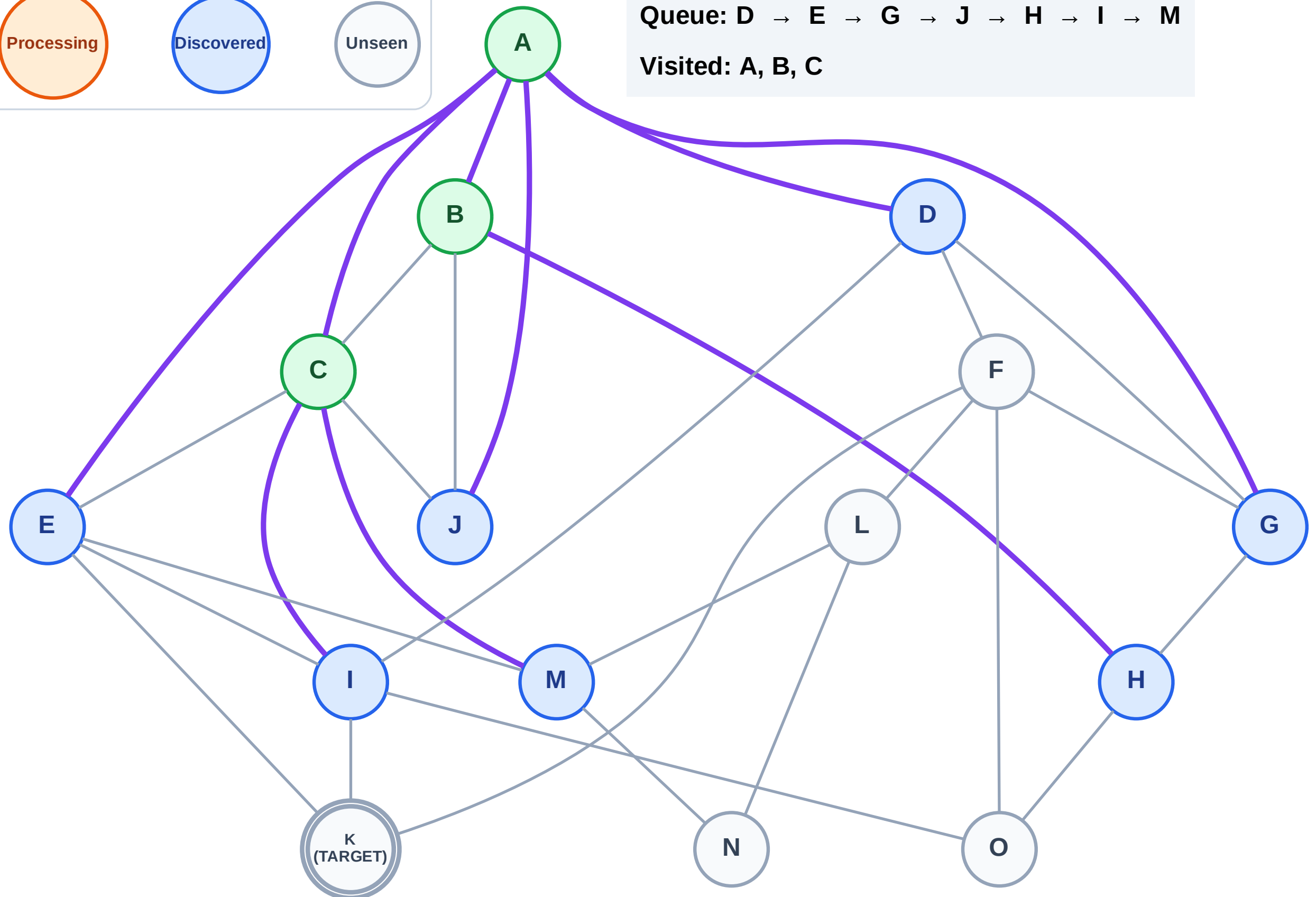
BFS Traversal

Step 7: Discover I, M from C



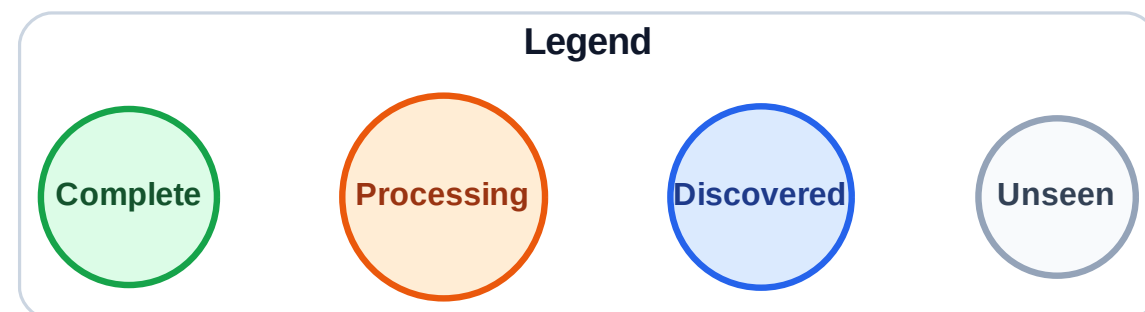
Queue: D → E → G → J → H → I → M

Visited: A, B, C



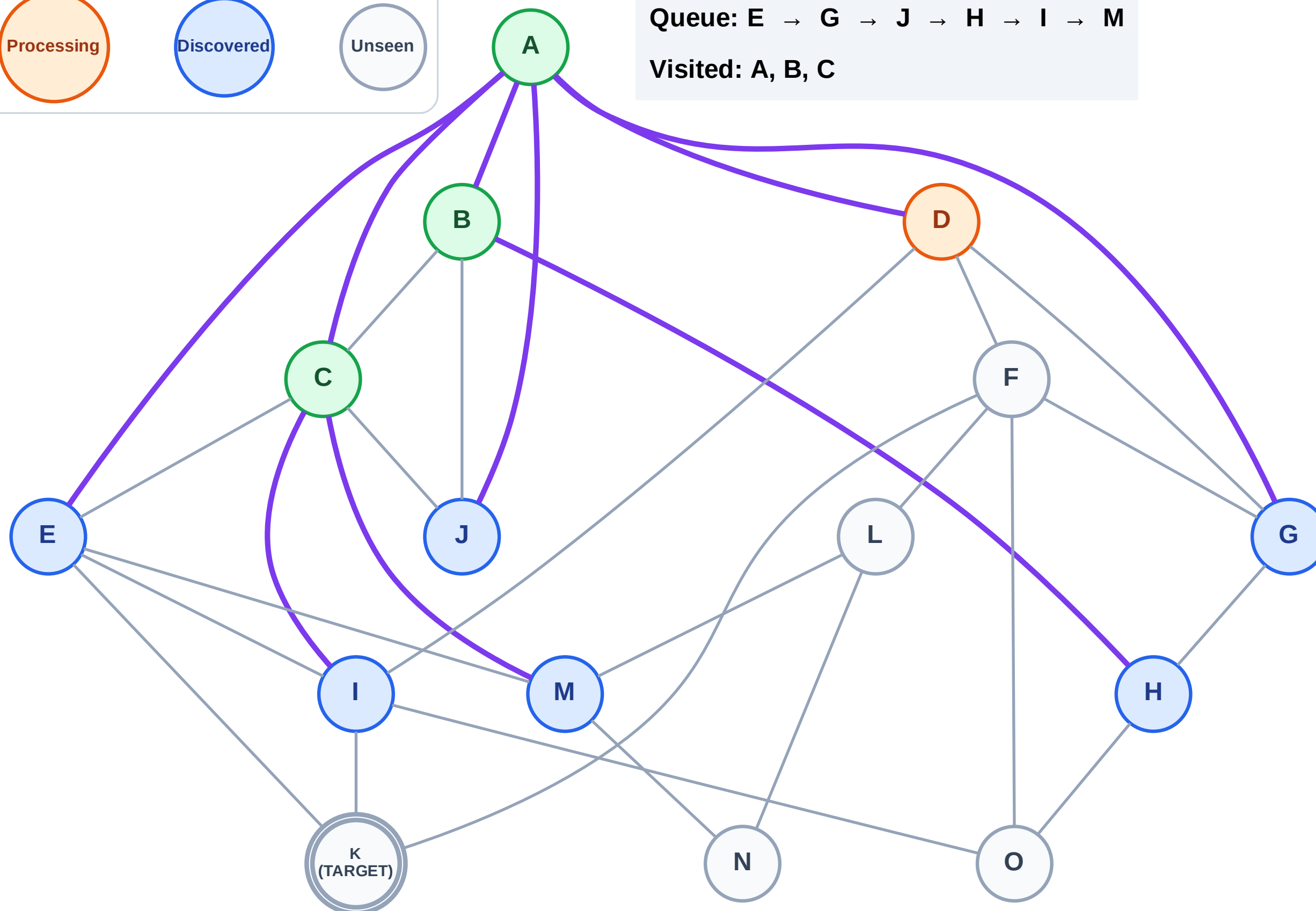
BFS Traversal

Step 8: Process node D



Queue: E → G → J → H → I → M

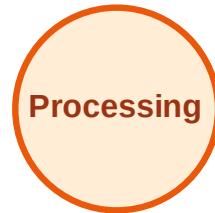
Visited: A, B, C



BFS Traversal

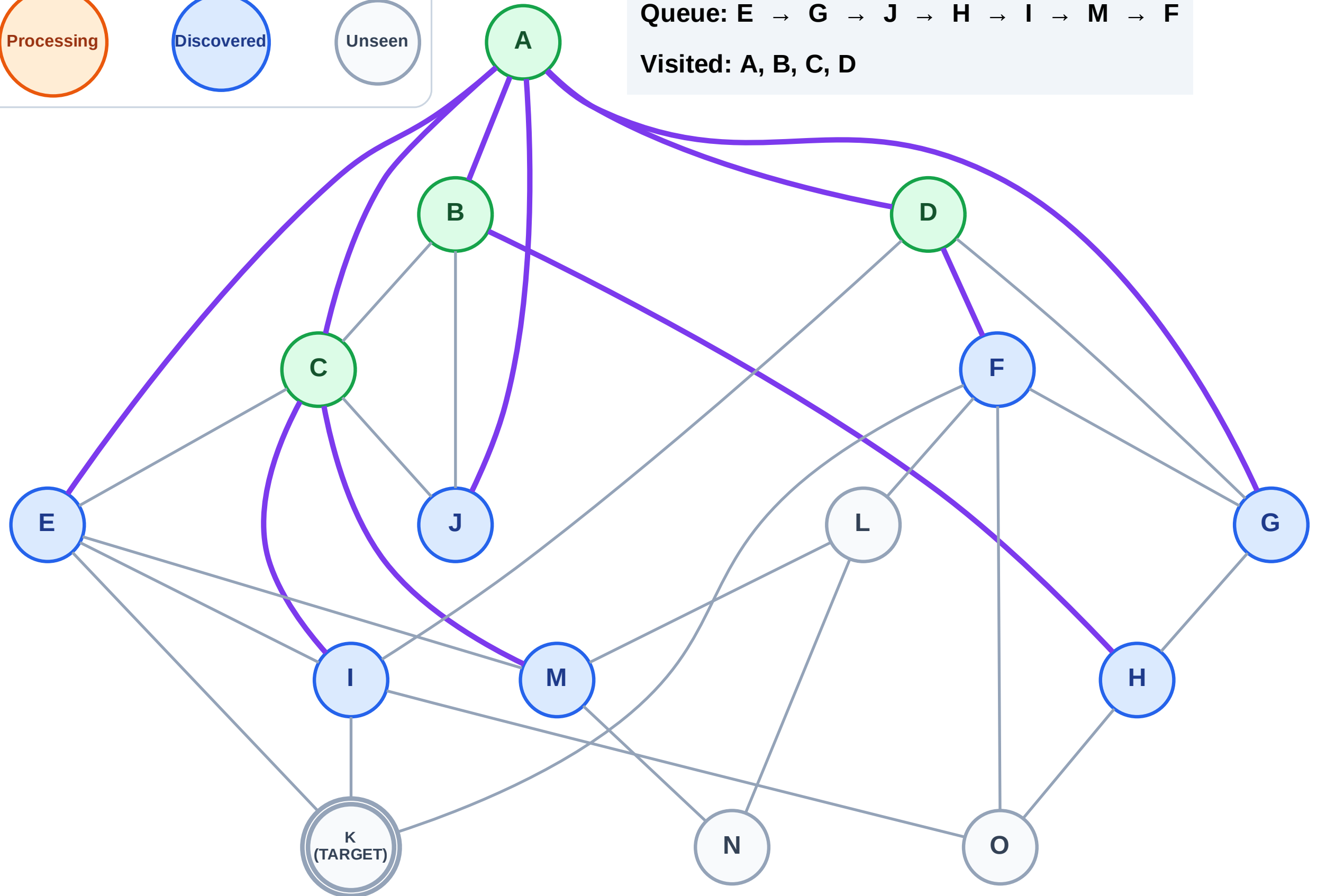
Step 9: Discover F from D

Legend



Queue: E → G → J → H → I → M → F

Visited: A, B, C, D



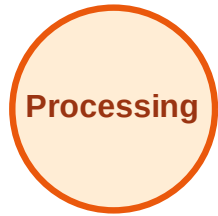
BFS Traversal

Step 10: Process node E

Legend



Complete



Processing



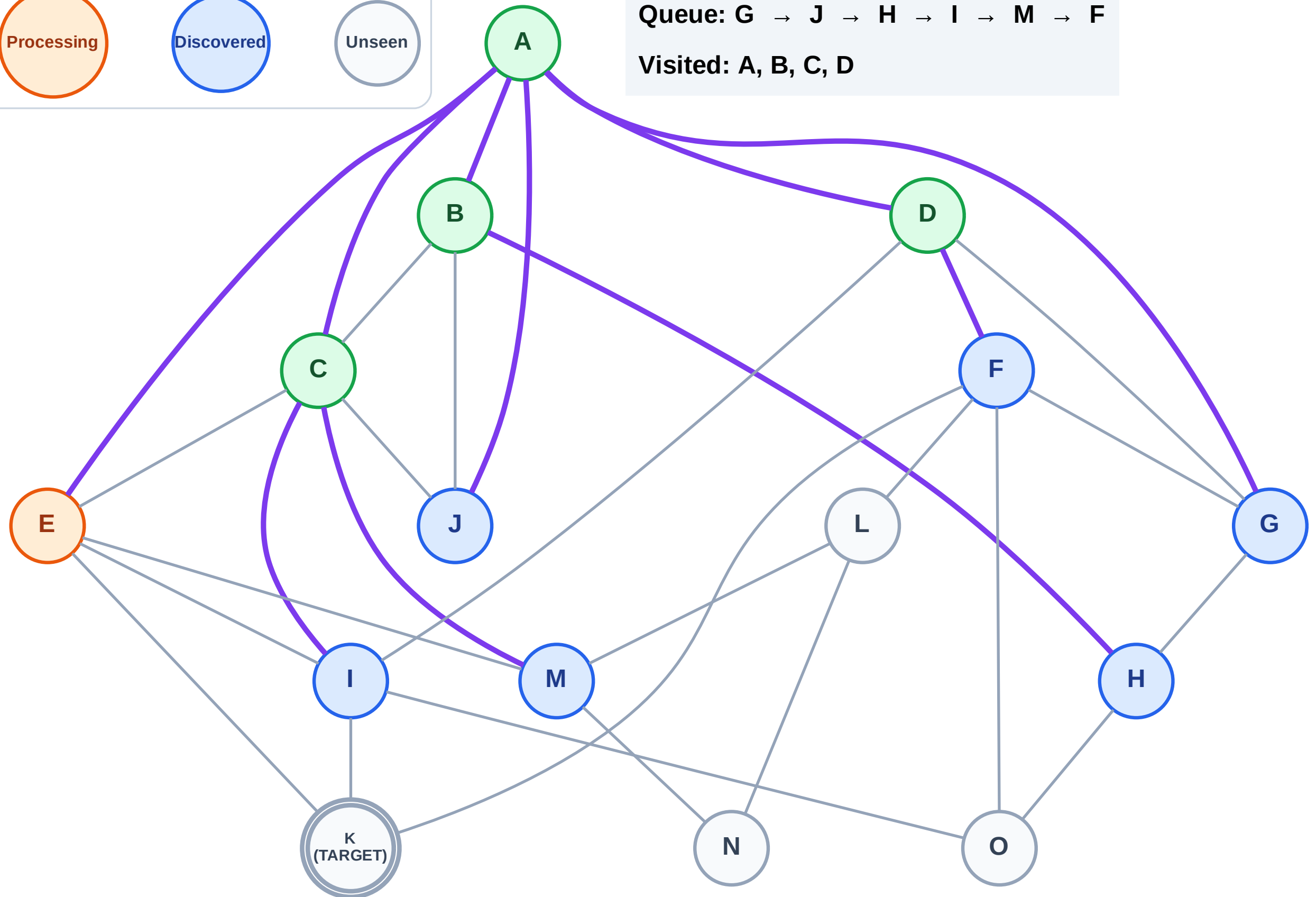
Discovered



Unseen

Queue: G → J → H → I → M → F

Visited: A, B, C, D



BFS Traversal

Step 11: Discover K from E

Legend

Complete

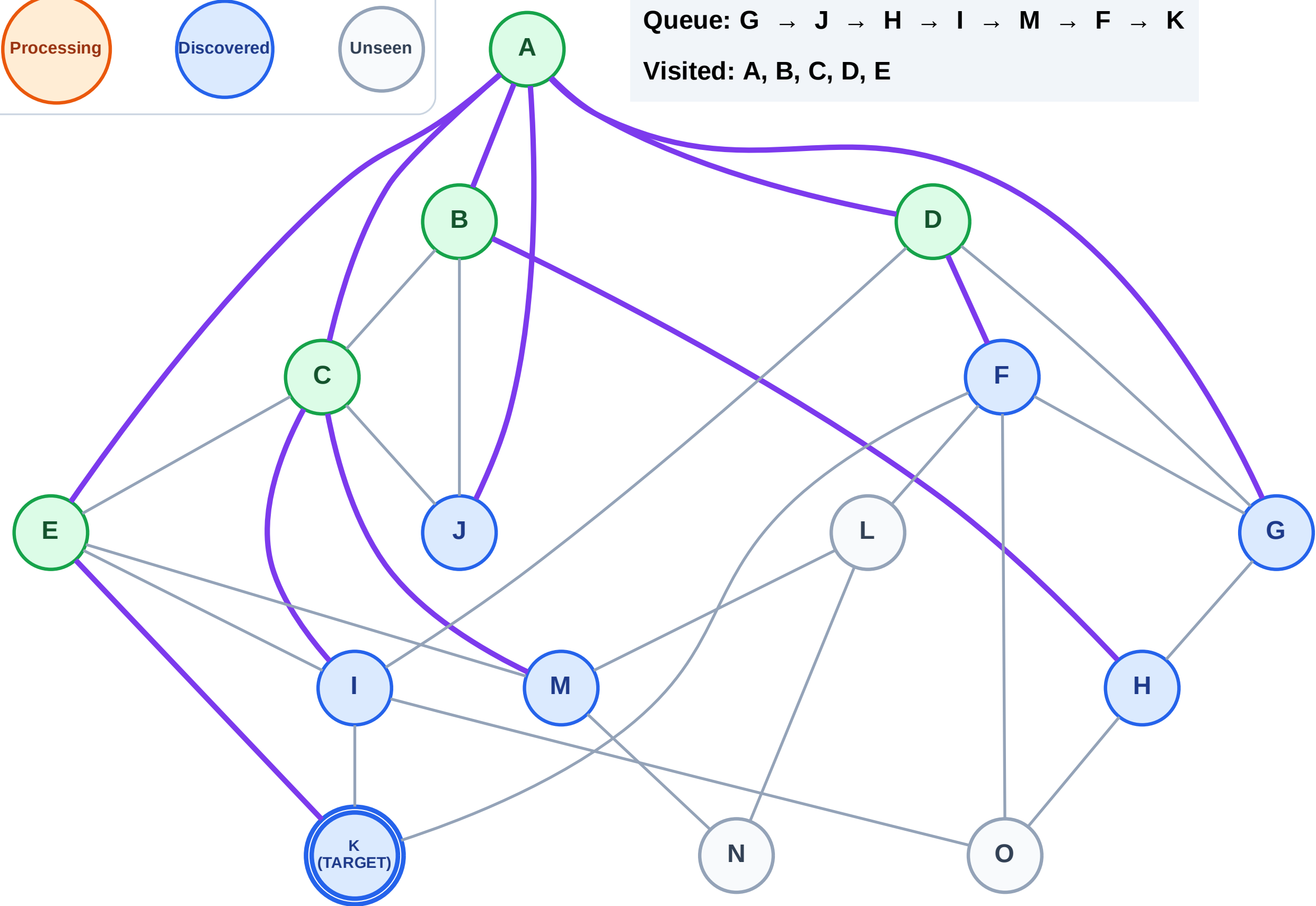
Processing

Discovered

Unseen

Queue: G → J → H → I → M → F → K

Visited: A, B, C, D, E



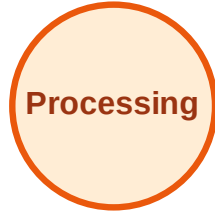
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



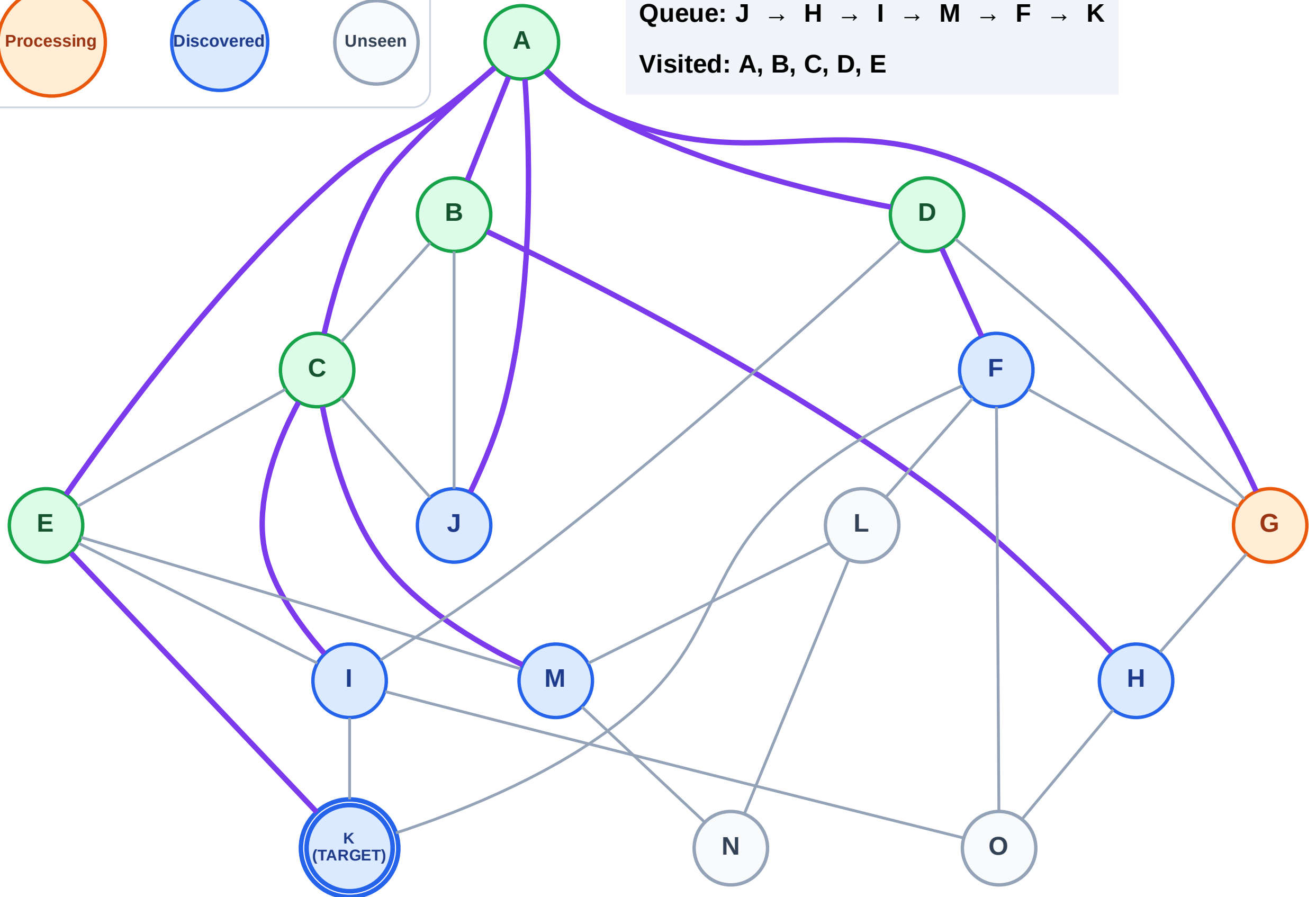
Discovered



Unseen

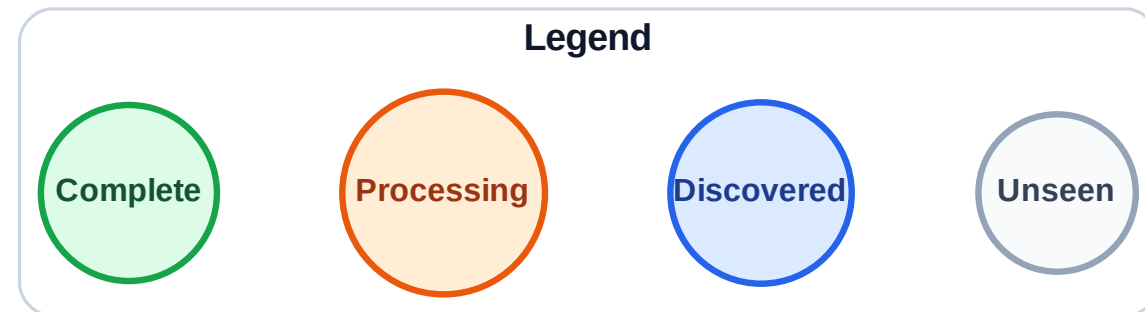
Queue: J → H → I → M → F → K

Visited: A, B, C, D, E

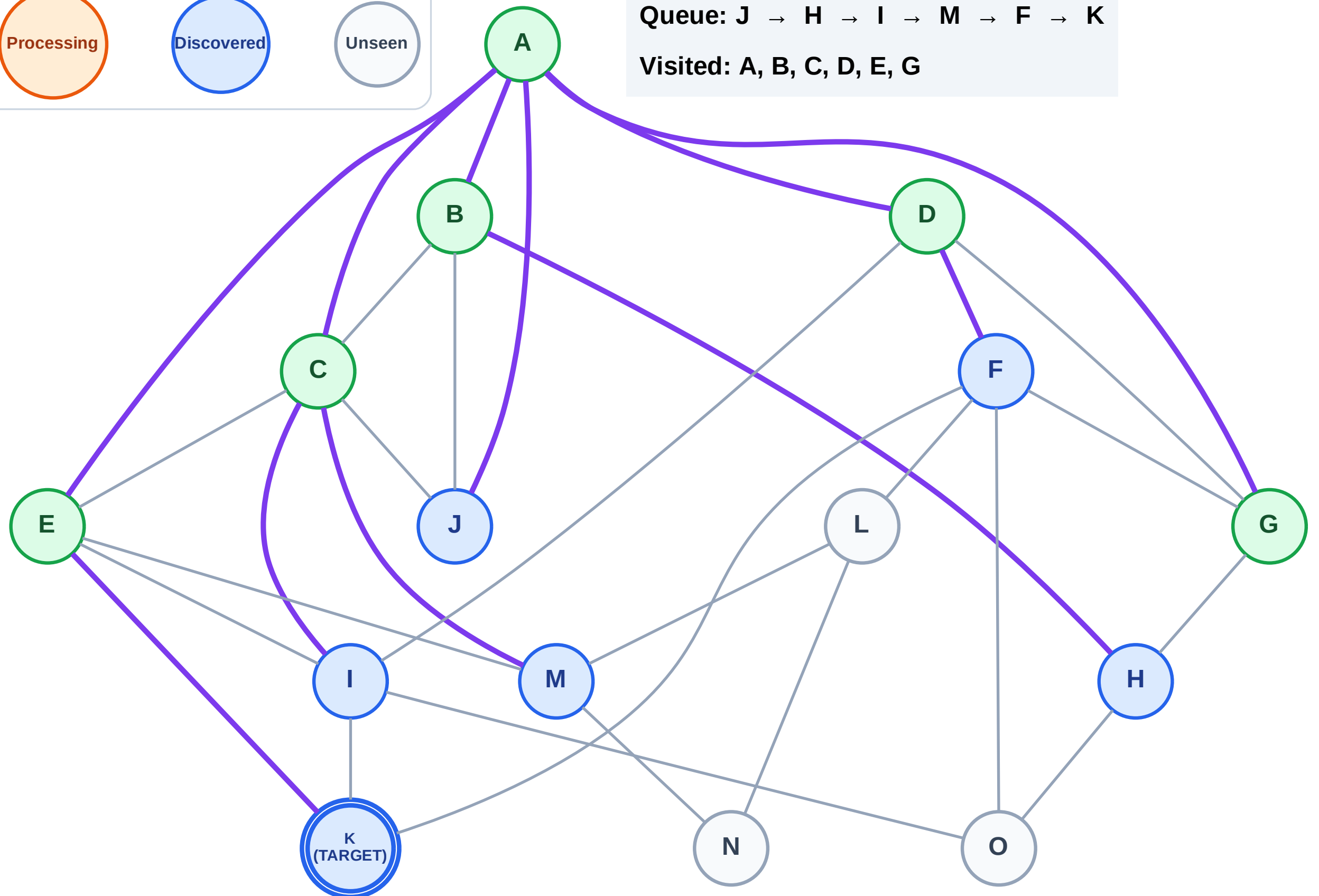


BFS Traversal

Step 13: No new neighbors from G

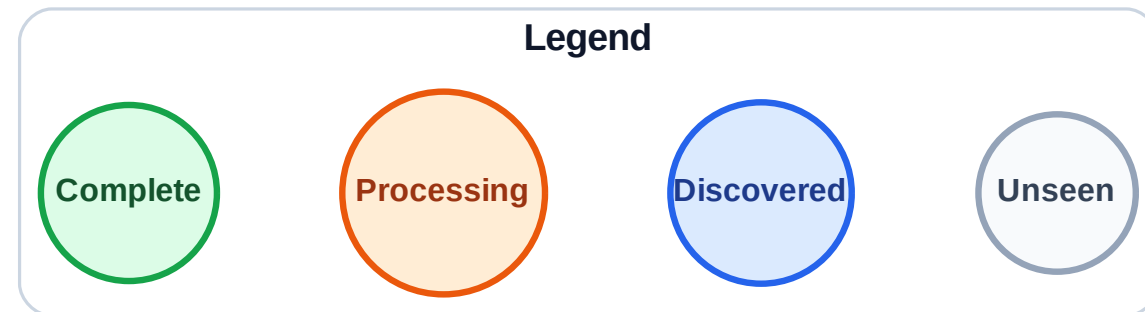


Queue: J → H → I → M → F → K
Visited: A, B, C, D, E, G



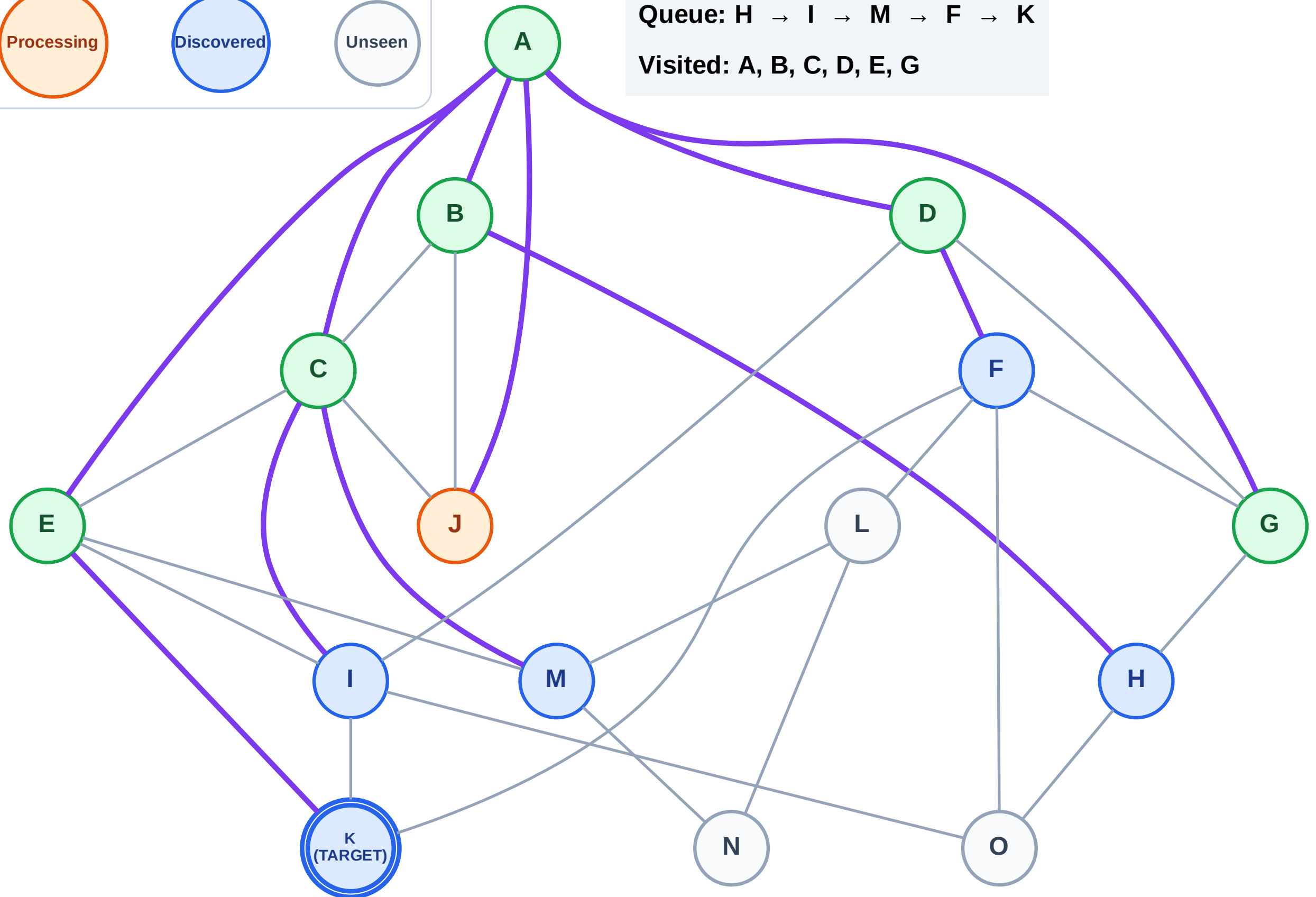
BFS Traversal

Step 14: Process node J



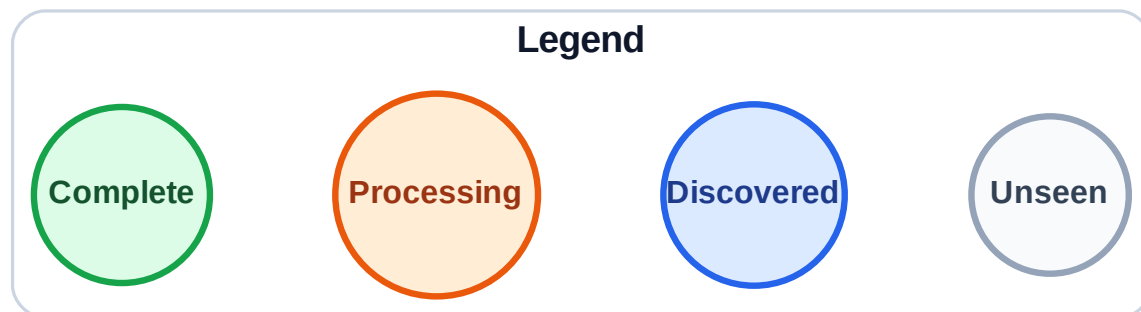
Queue: H → I → M → F → K

Visited: A, B, C, D, E, G

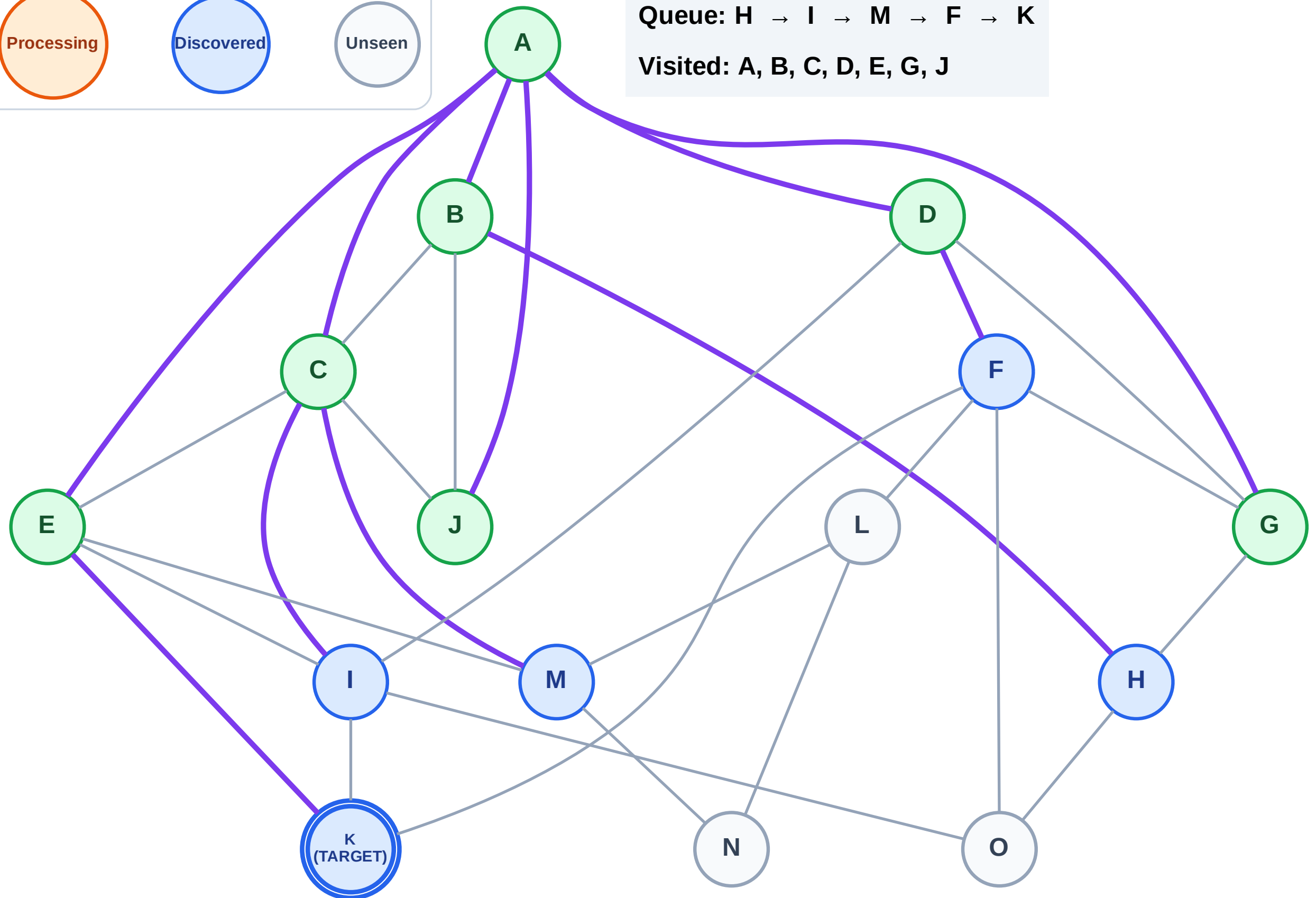


BFS Traversal

Step 15: No new neighbors from J

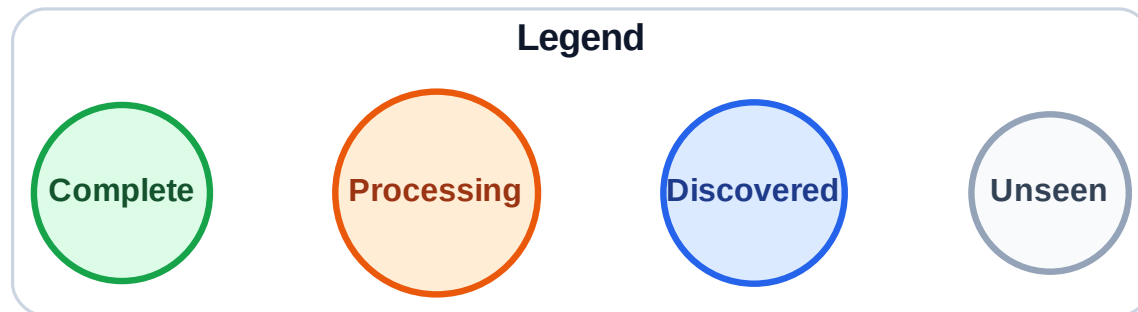


Queue: H → I → M → F → K
Visited: A, B, C, D, E, G, J

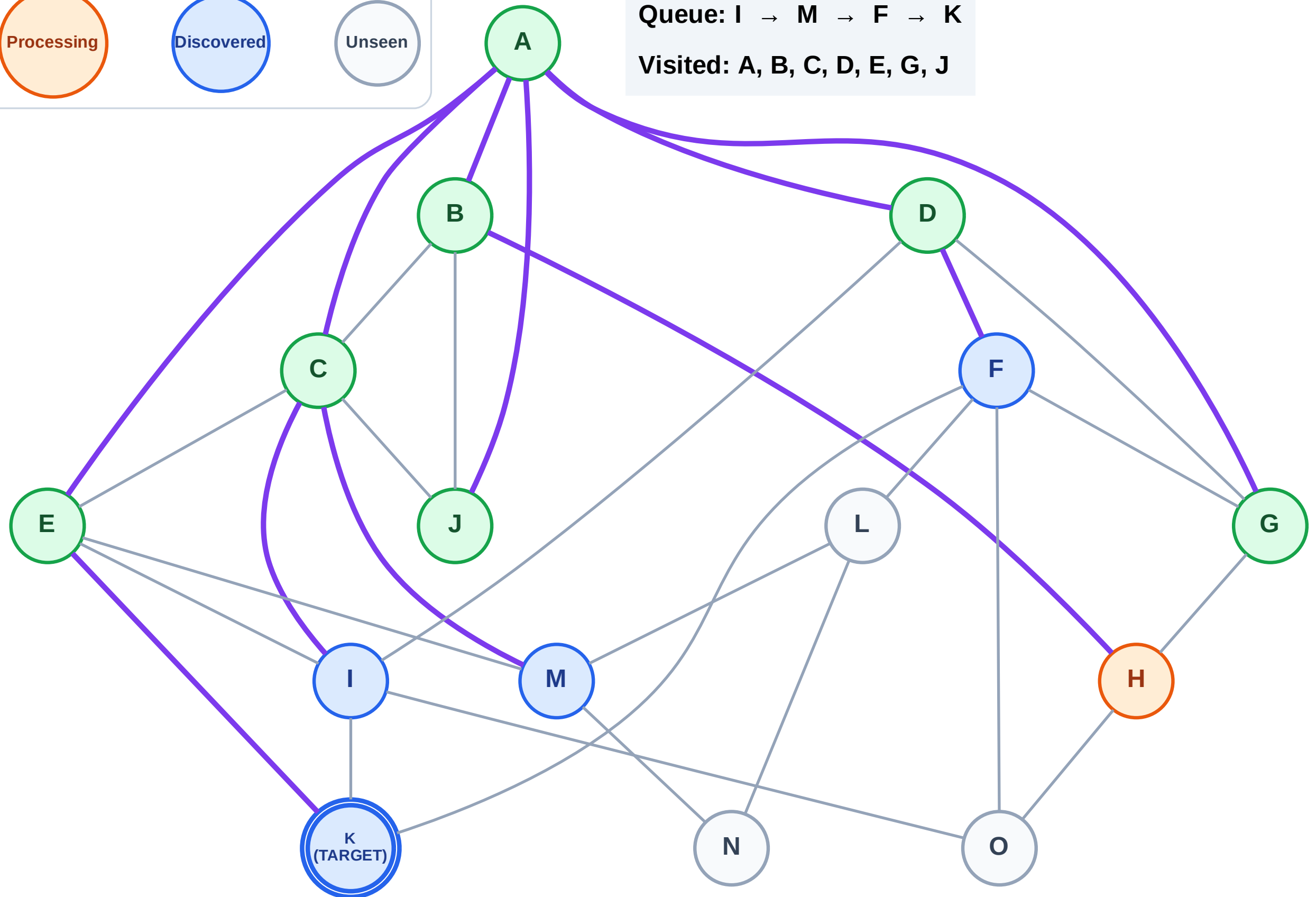


BFS Traversal

Step 16: Process node H



Queue: I → M → F → K
Visited: A, B, C, D, E, G, J



BFS Traversal

Step 17: Discover O from H

Legend

Complete

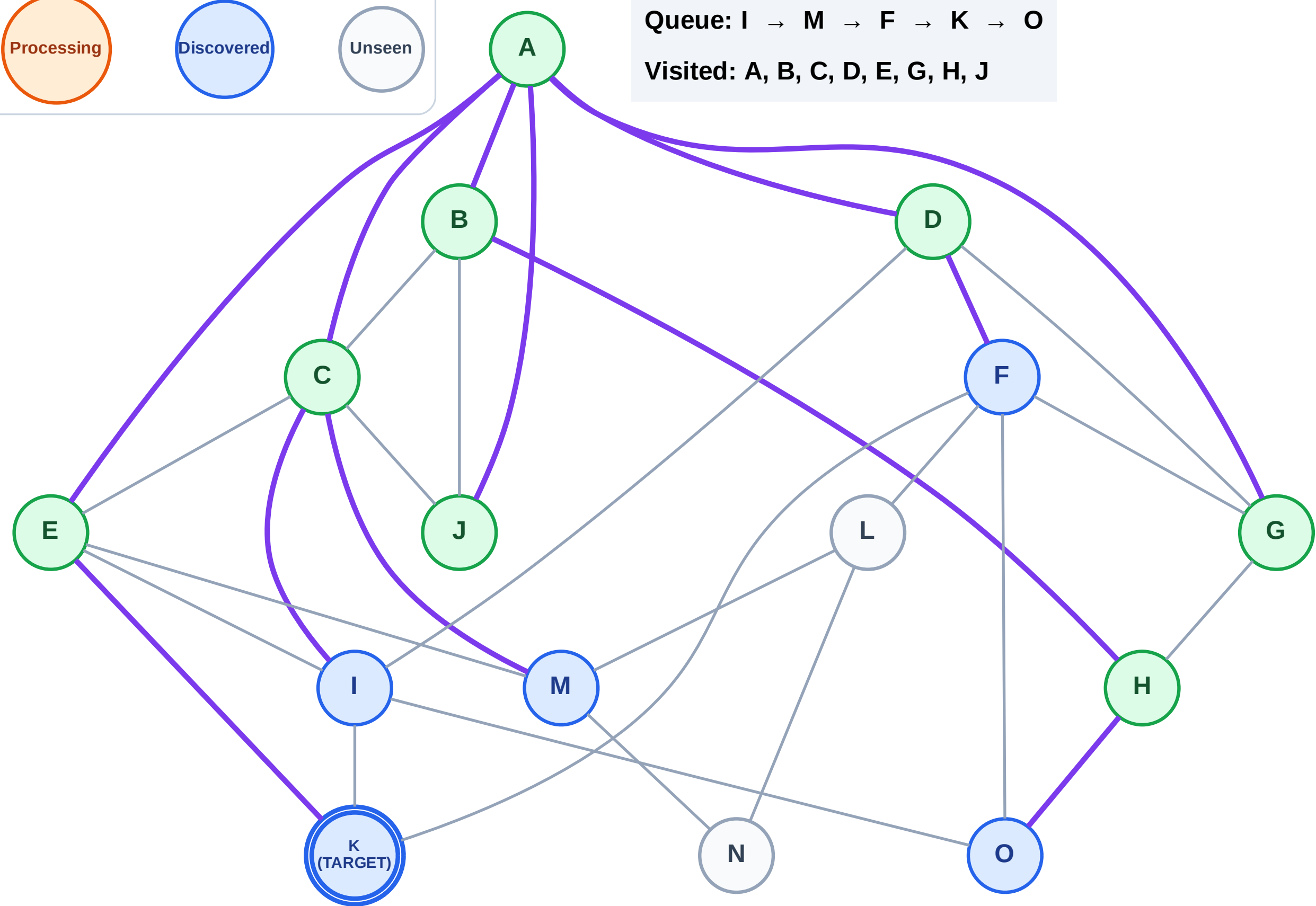
Processing

Discovered

Unseen

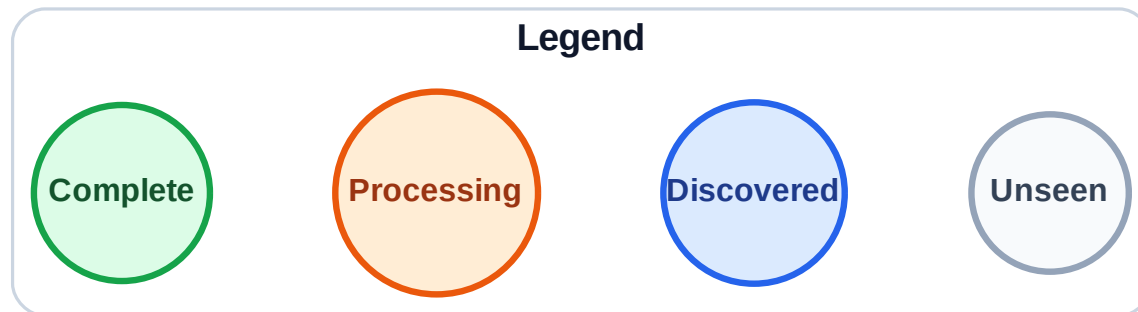
Queue: I → M → F → K → O

Visited: A, B, C, D, E, G, H, J

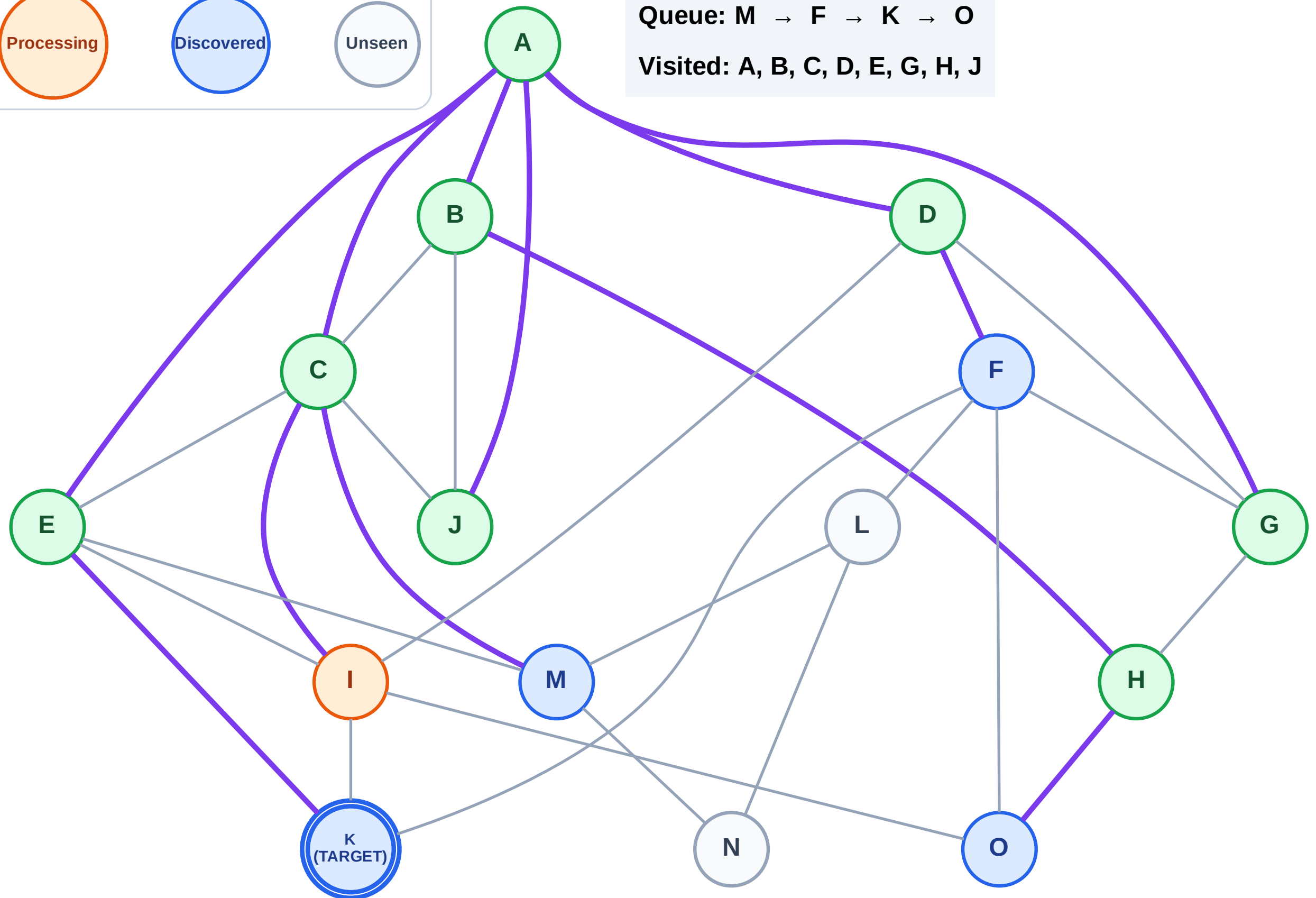


BFS Traversal

Step 18: Process node I

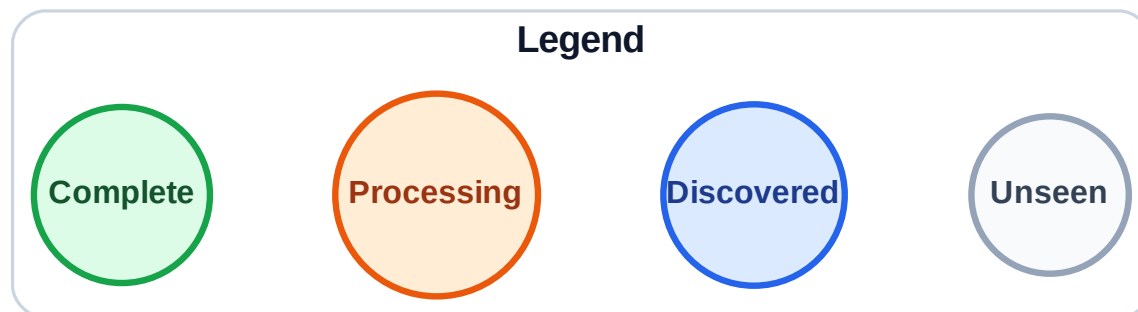


Queue: M → F → K → O
Visited: A, B, C, D, E, G, H, J

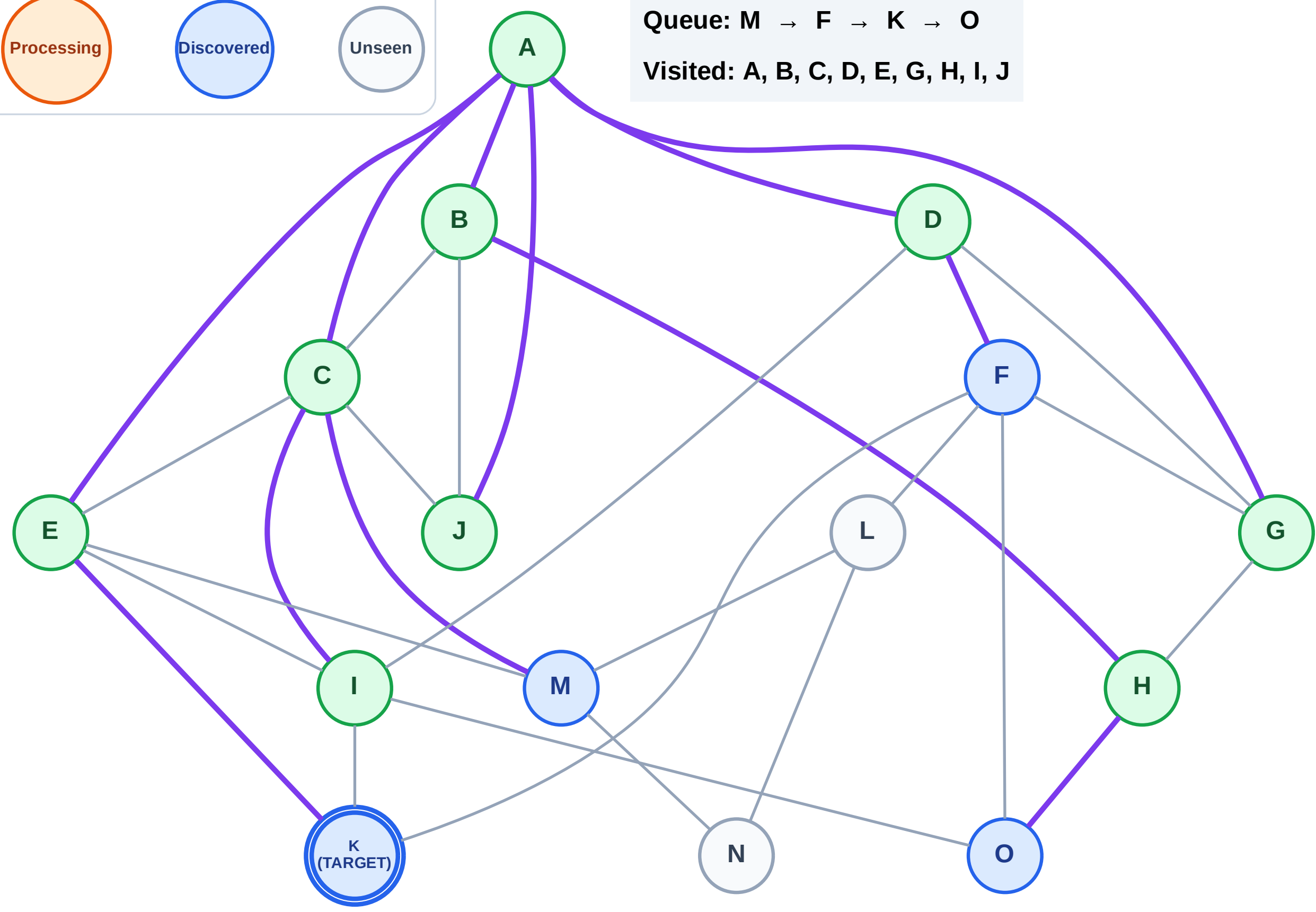


BFS Traversal

Step 19: No new neighbors from I



Queue: M → F → K → O
Visited: A, B, C, D, E, G, H, I, J



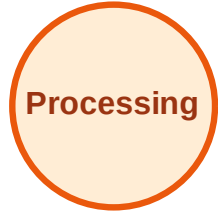
BFS Traversal

Step 20: Process node M

Legend



Complete



Processing



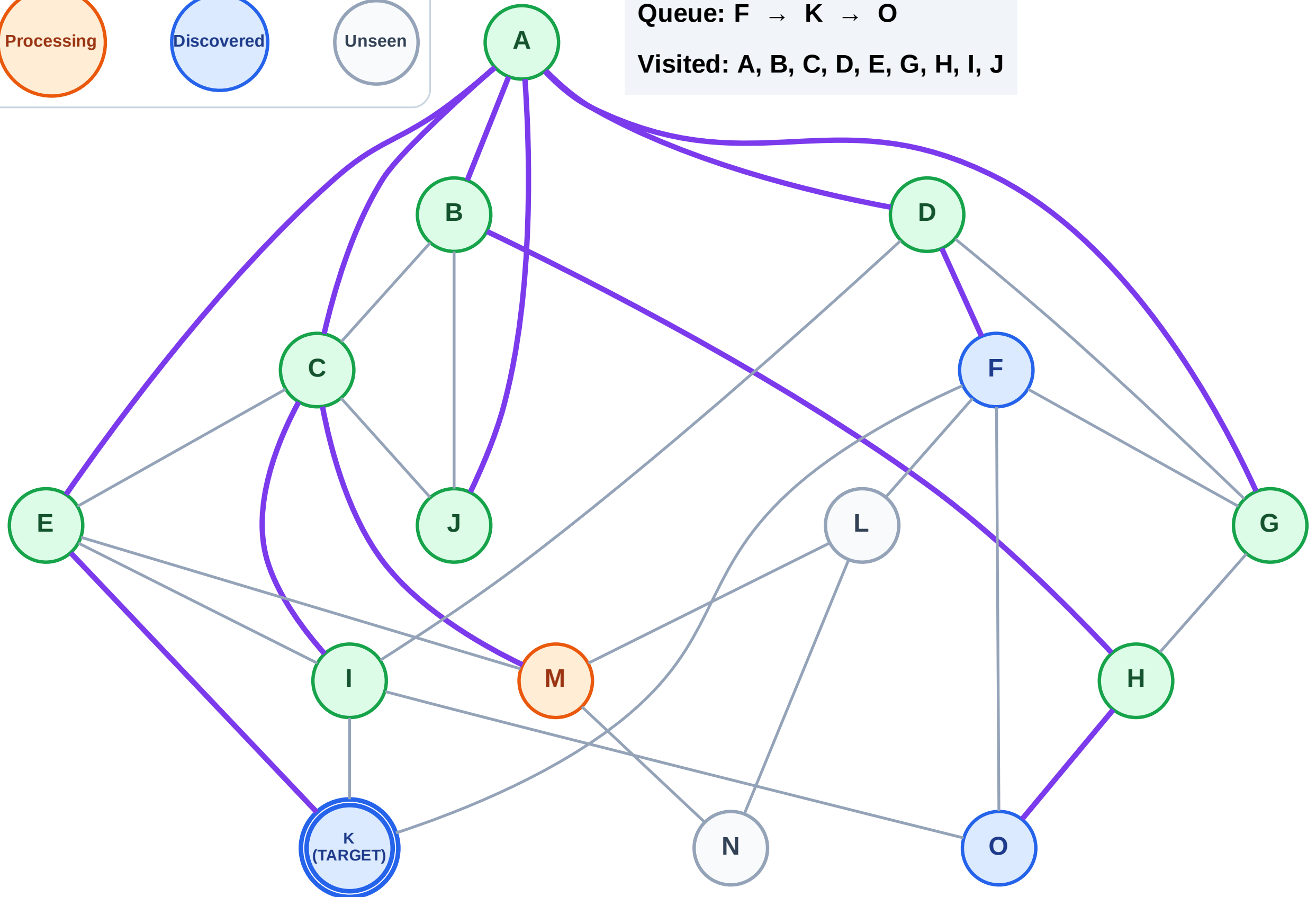
Discovered



Unseen

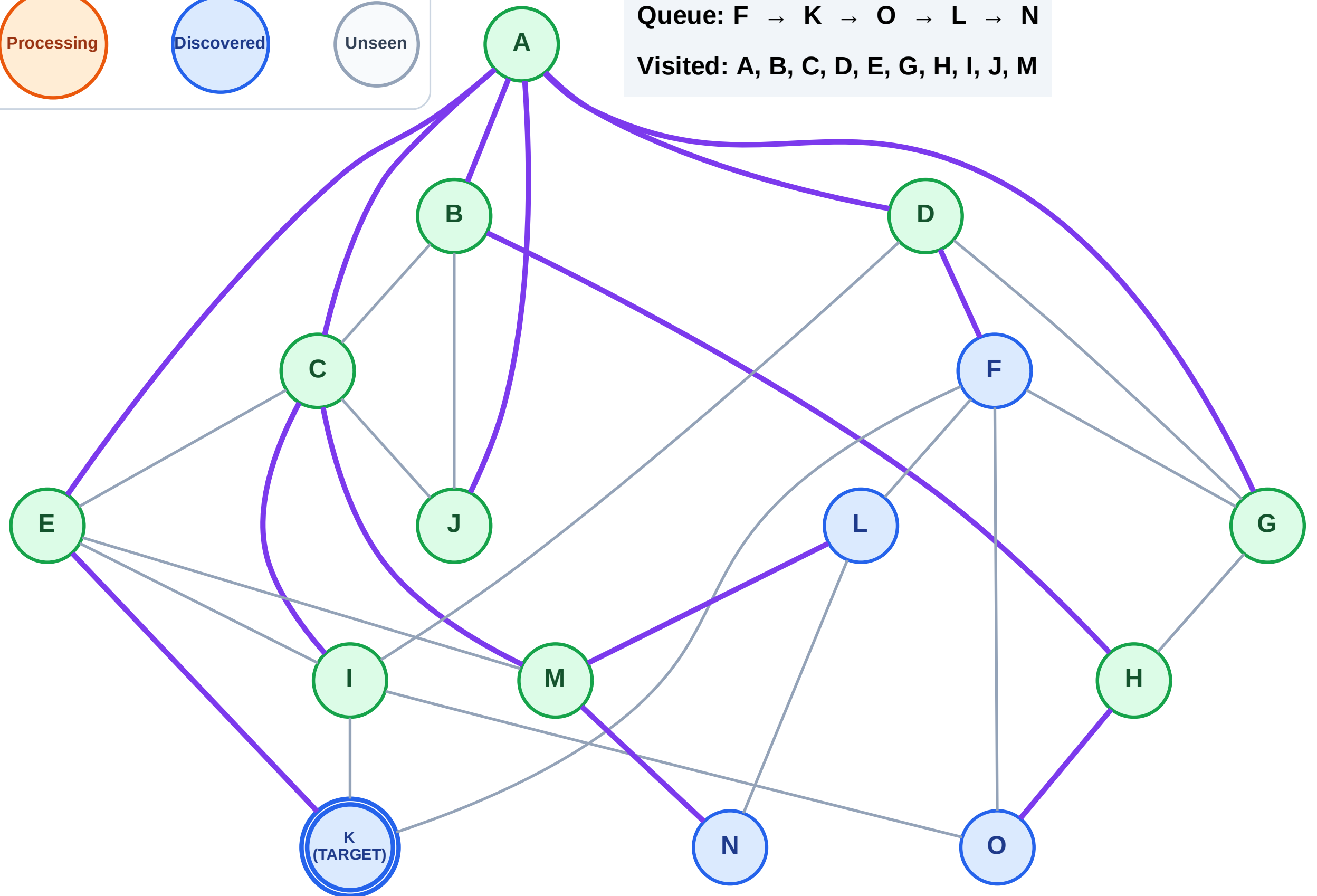
Queue: F → K → O

Visited: A, B, C, D, E, G, H, I, J



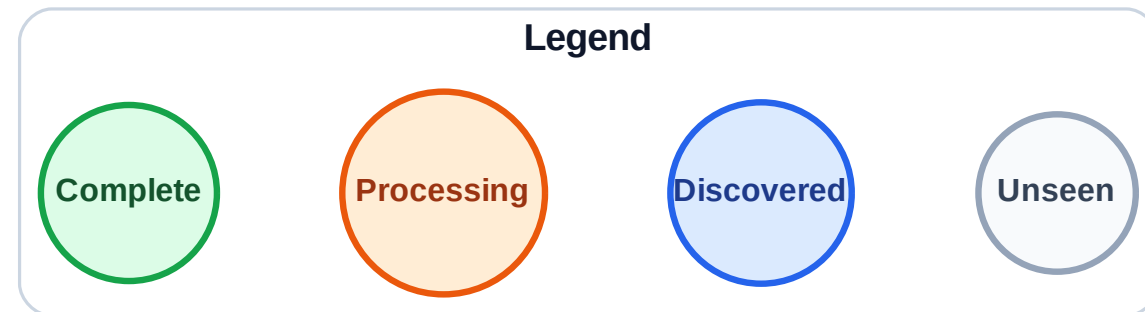
Step 21: Discover L, N from M

Queue: F → K → O → L → N
Visited: A, B, C, D, E, G, H, I, J, M



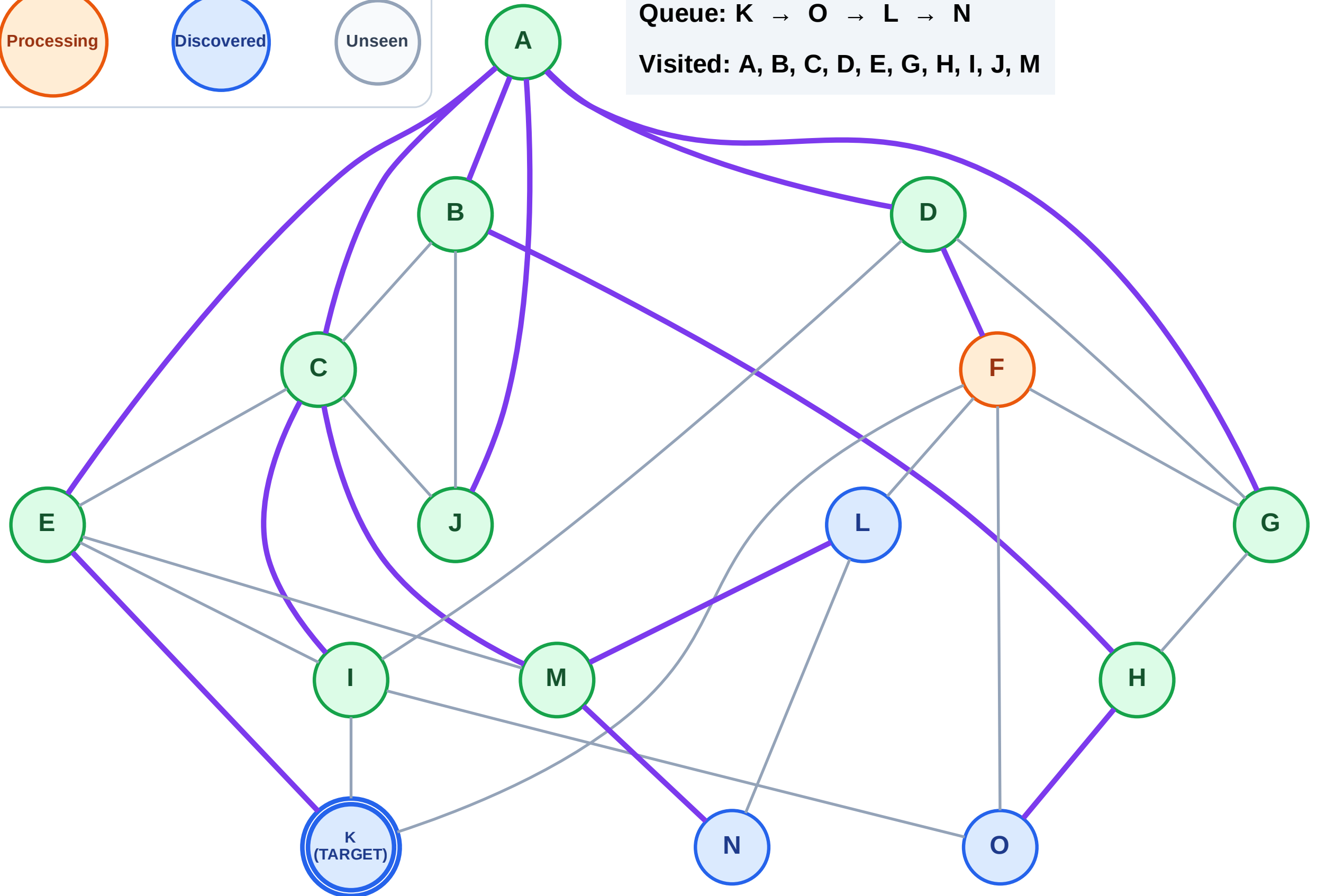
BFS Traversal

Step 22: Process node F



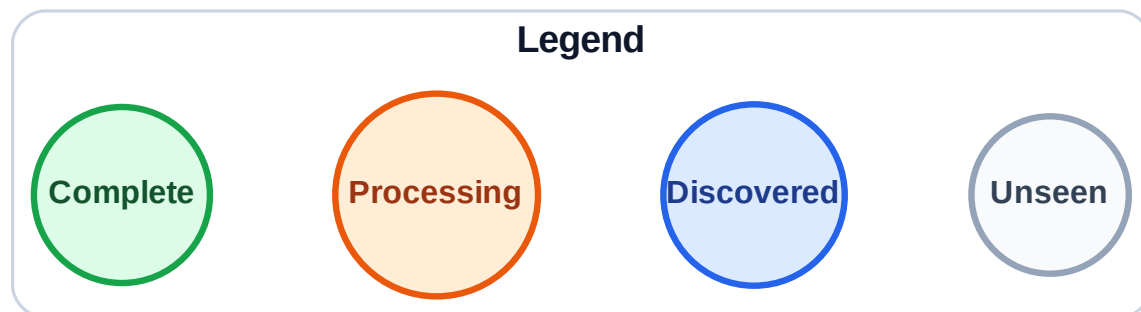
Queue: K → O → L → N

Visited: A, B, C, D, E, G, H, I, J, M



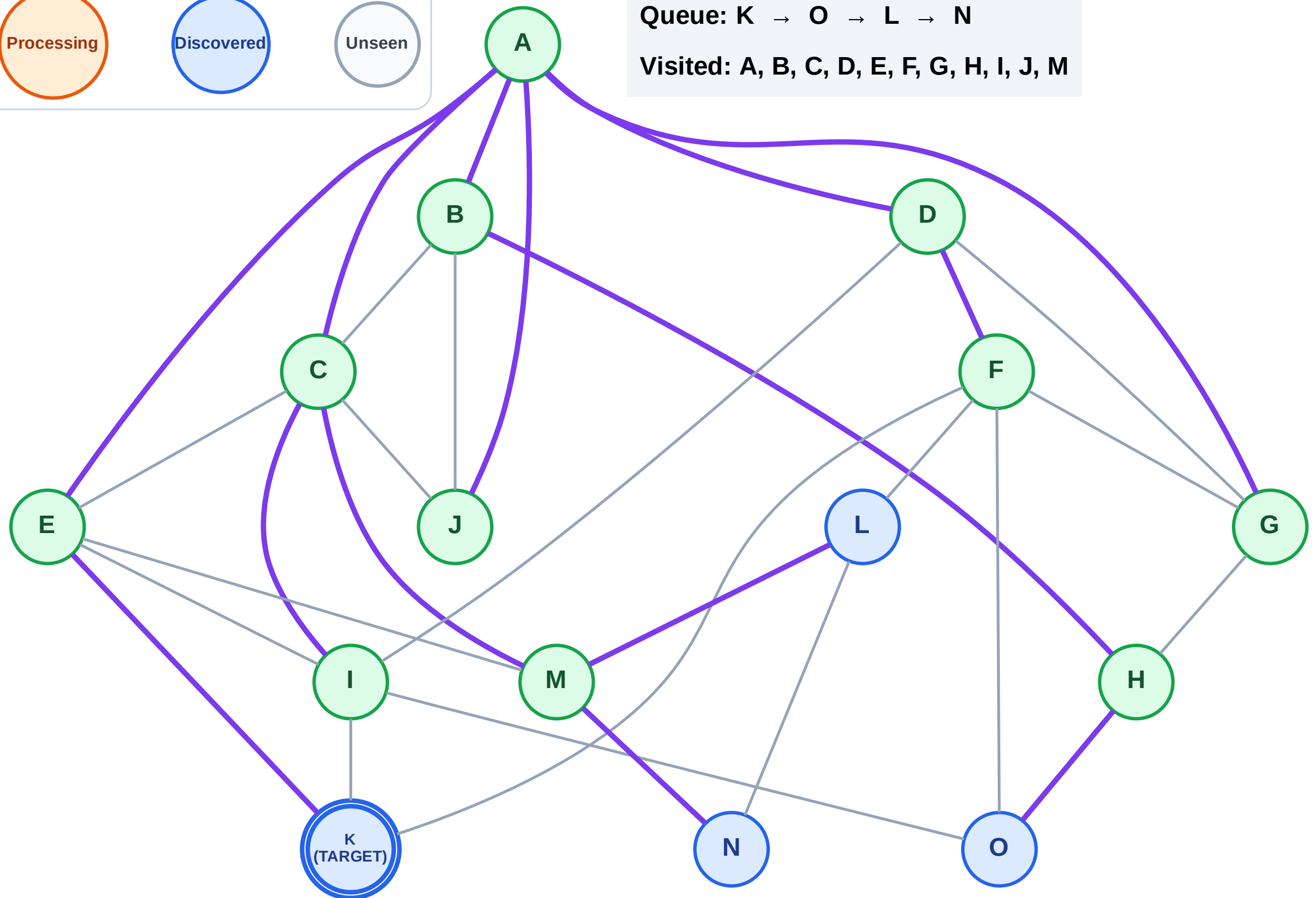
BFS Traversal

Step 23: No new neighbors from F



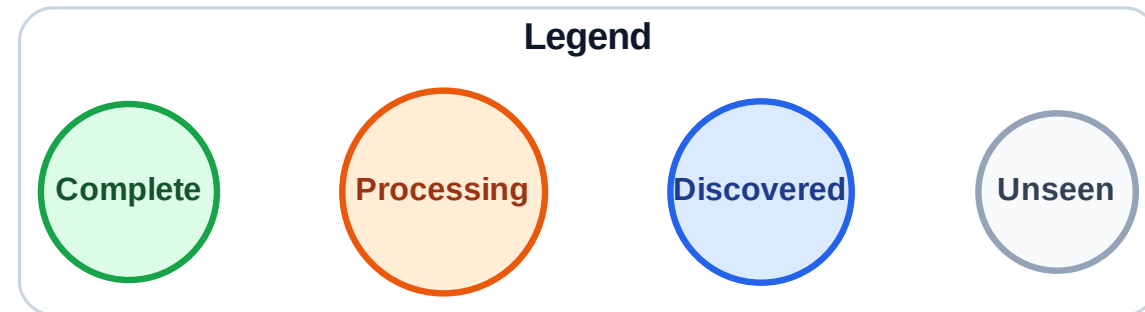
Queue: K → O → L → N

Visited: A, B, C, D, E, F, G, H, I, J, M



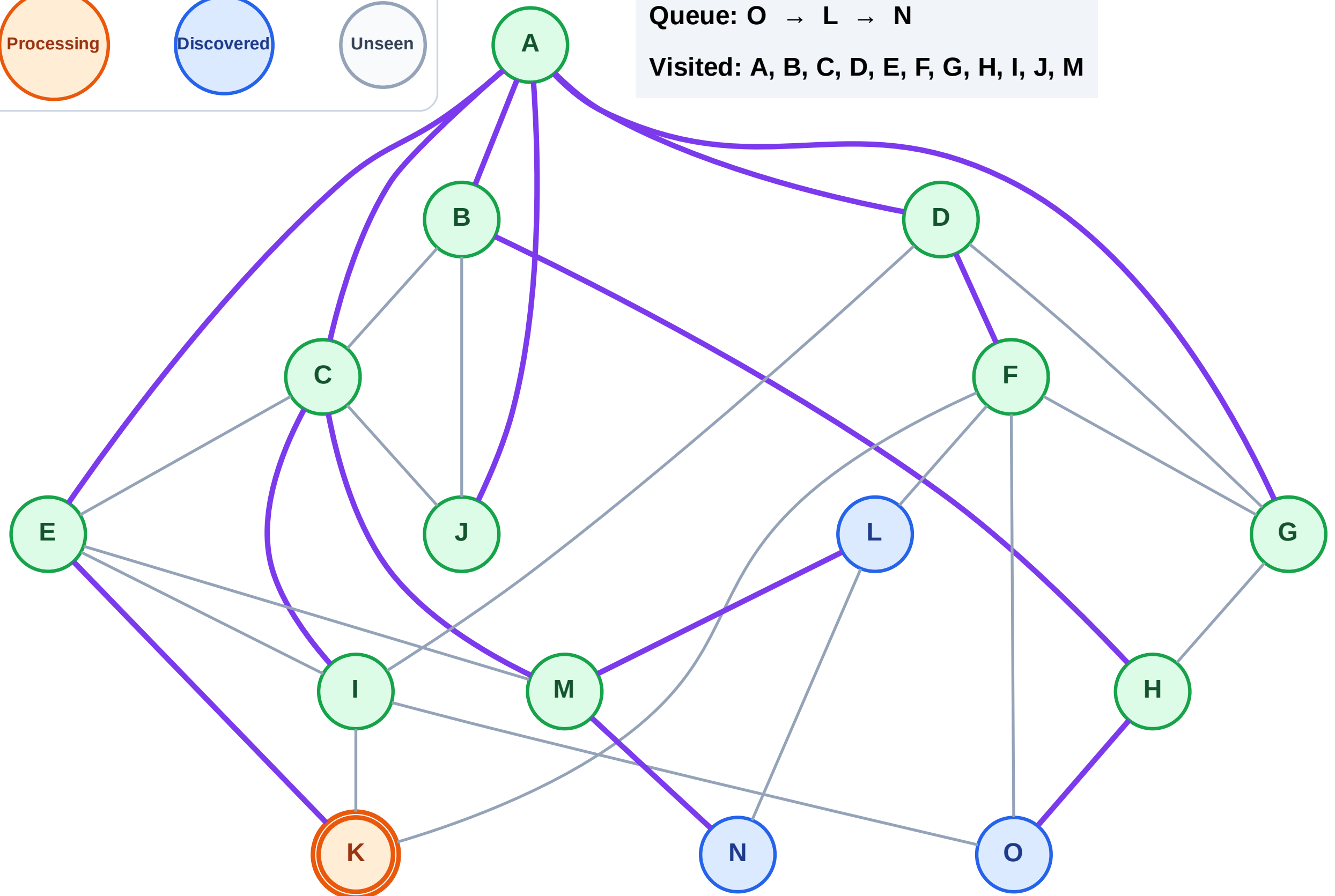
BFS Traversal

Step 24: Process node K



Queue: O → L → N

Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

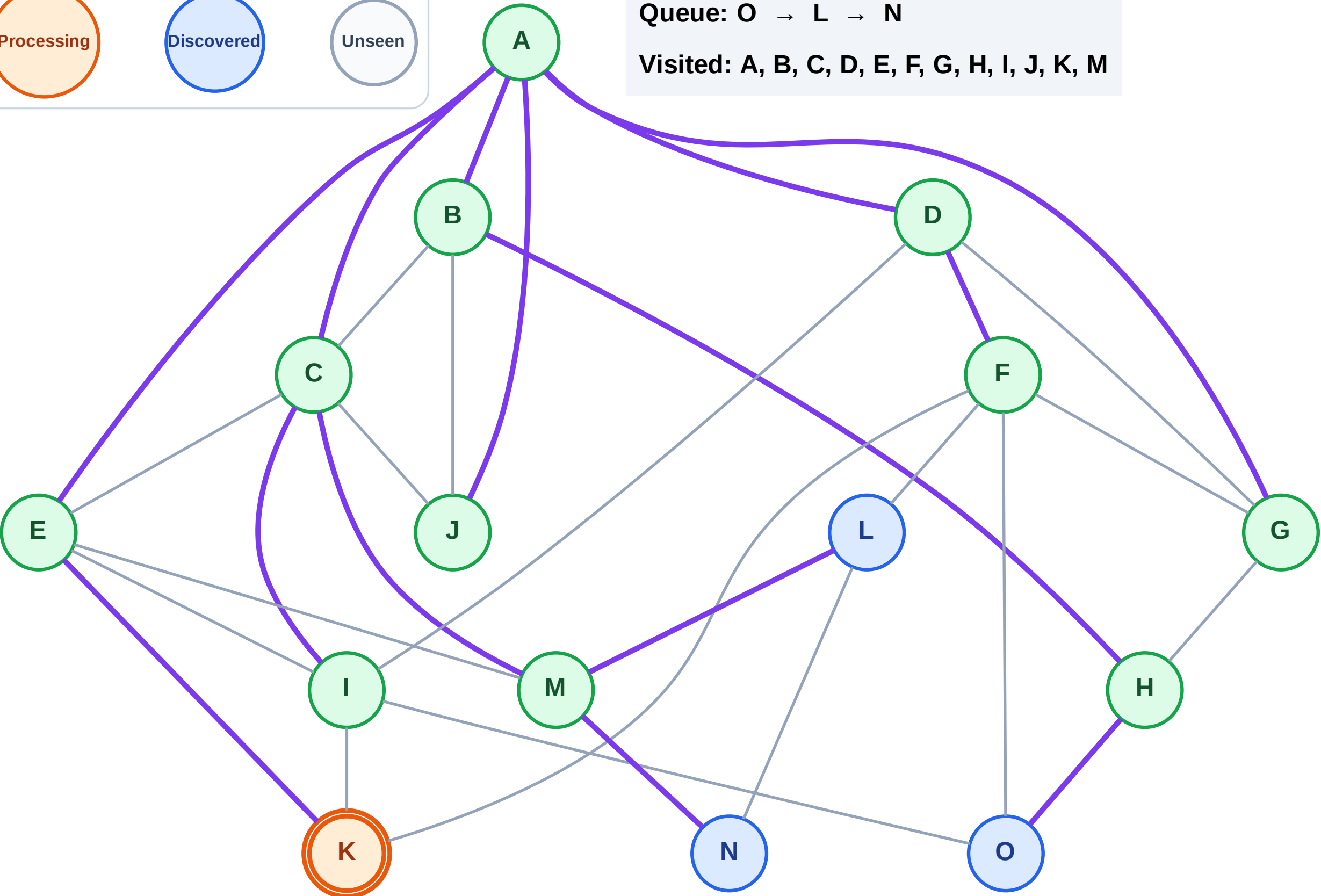
Processing

Discovered

Unseen

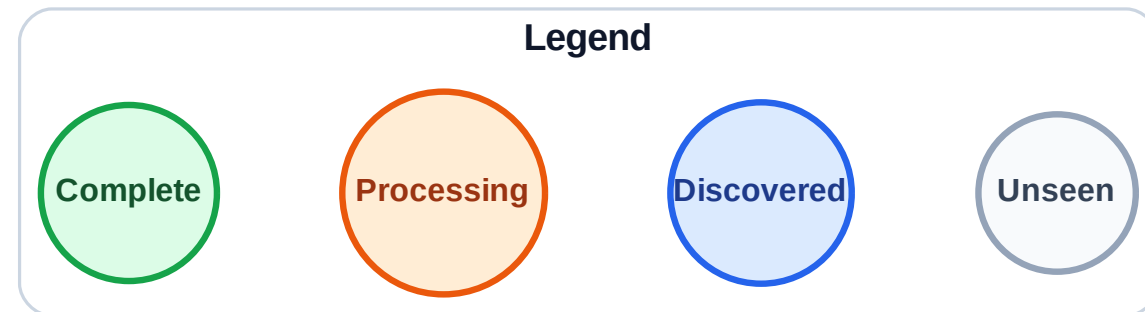
Queue: O → L → N

Visited: A, B, C, D, E, F, G, H, I, J, K, M

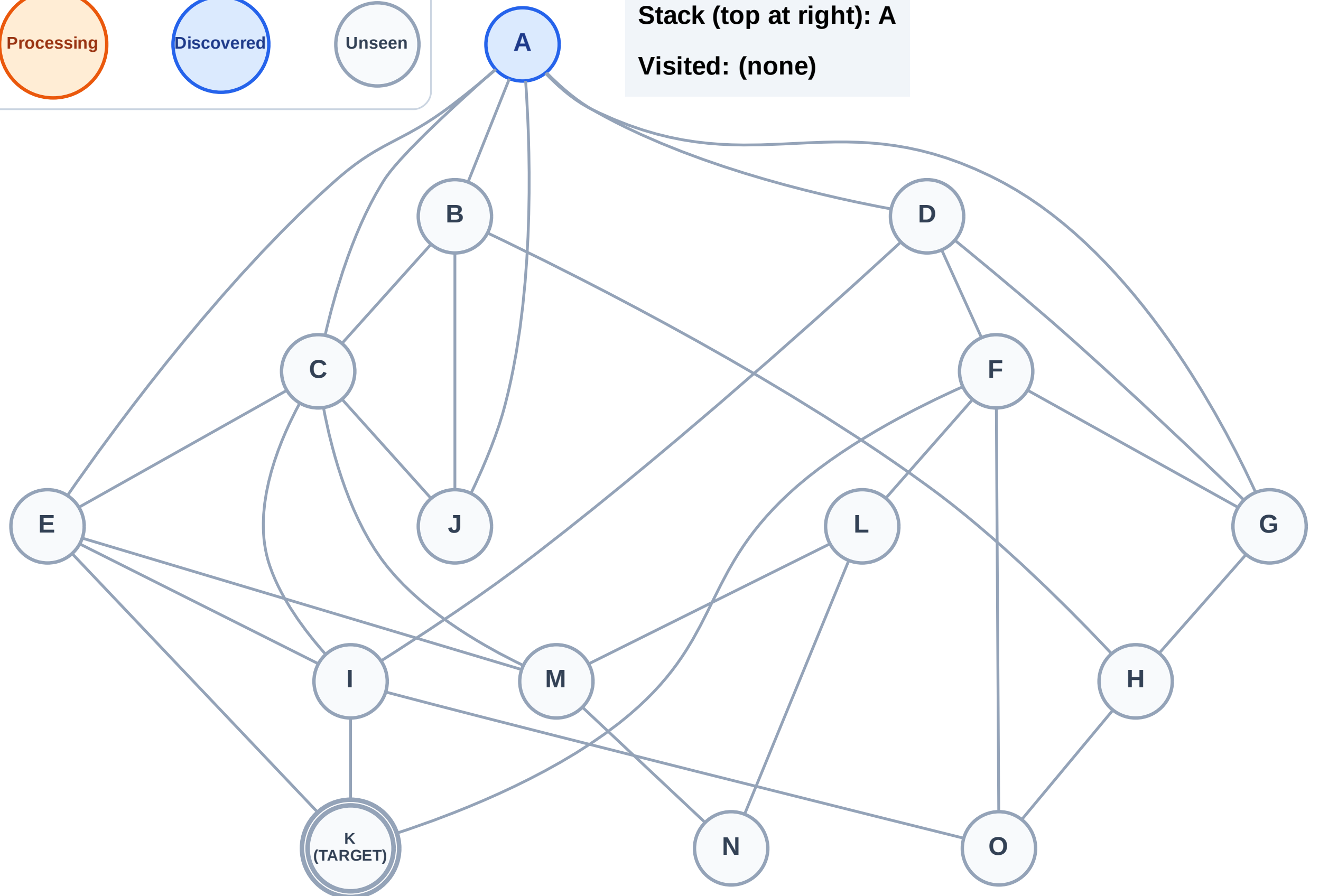


DFS Traversal

Step 1: Start at A (Seeking target node K)

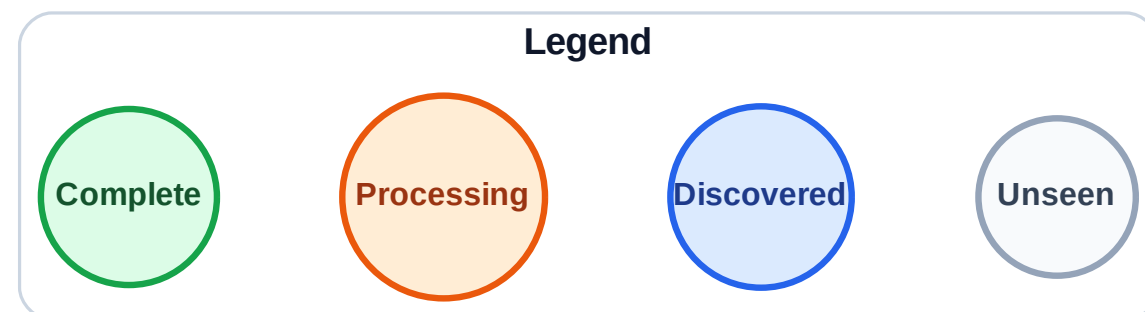


Stack (top at right): A
Visited: (none)

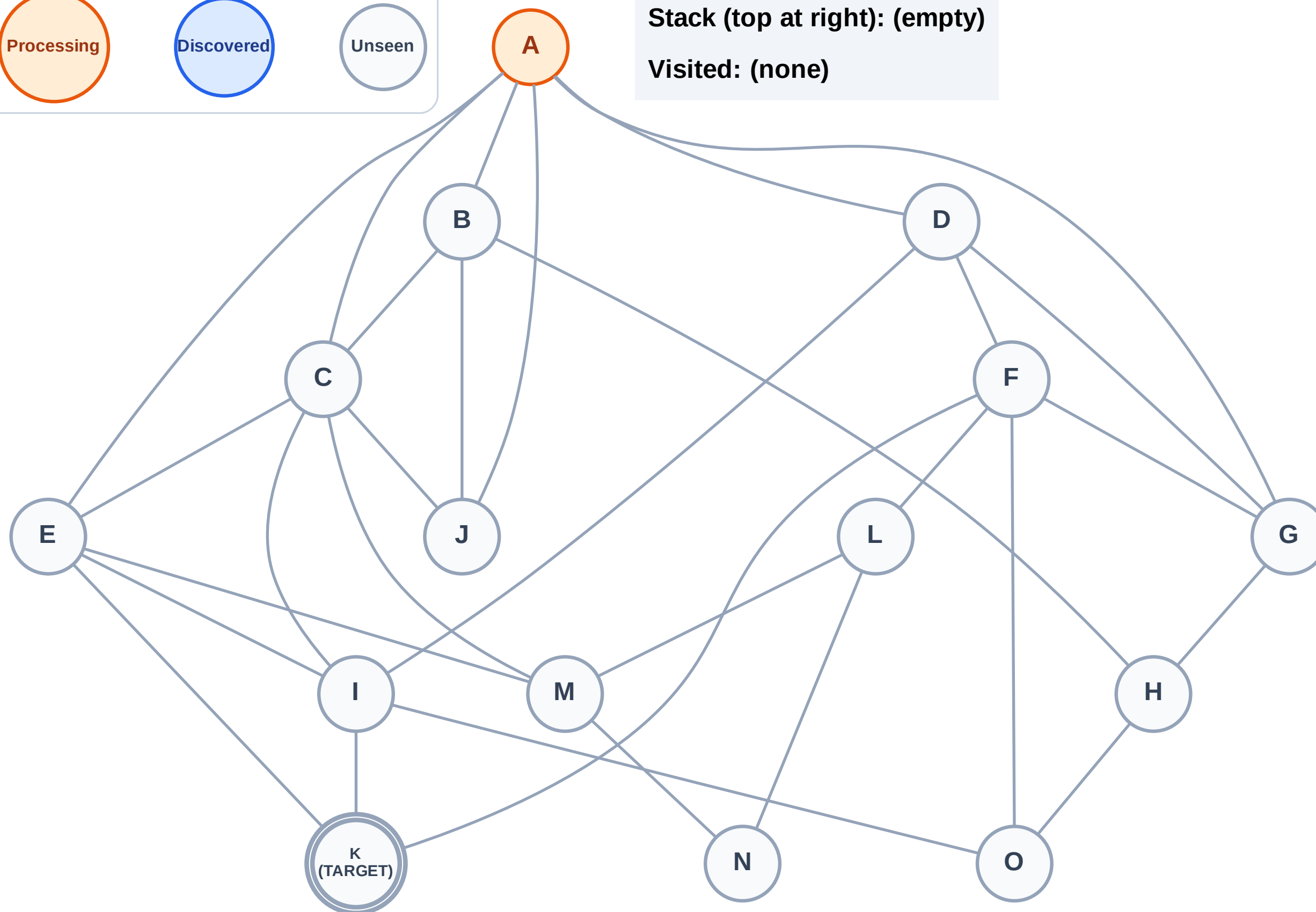


DFS Traversal

Step 2: Process node A



Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover J, G, E, D, C, B from A

Legend

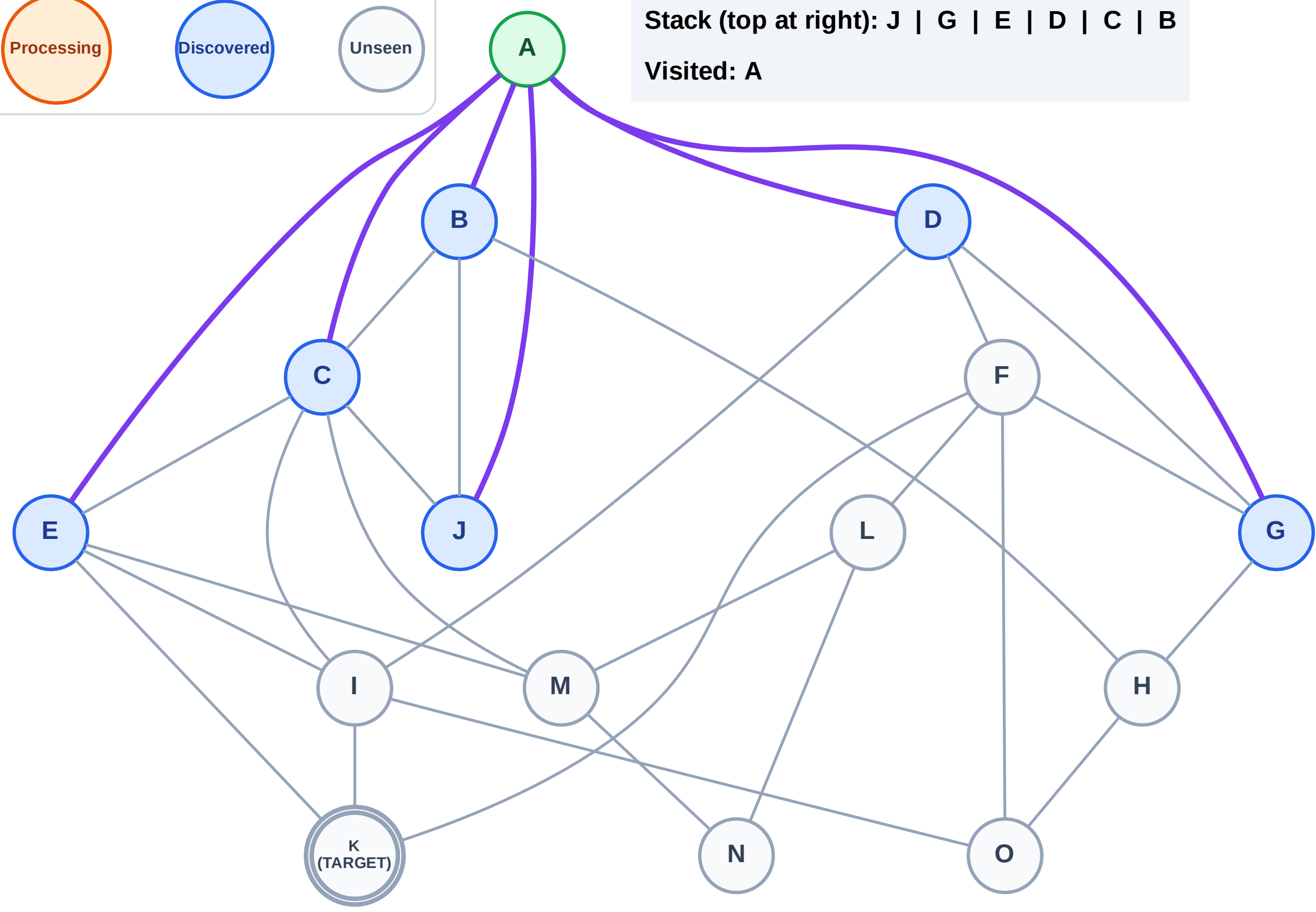
Complete

Processing

Discovered

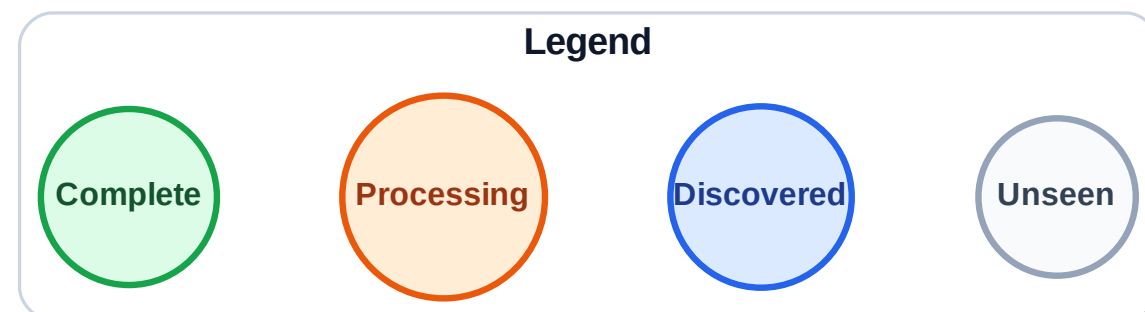
Unseen

Stack (top at right): J | G | E | D | C | B
Visited: A

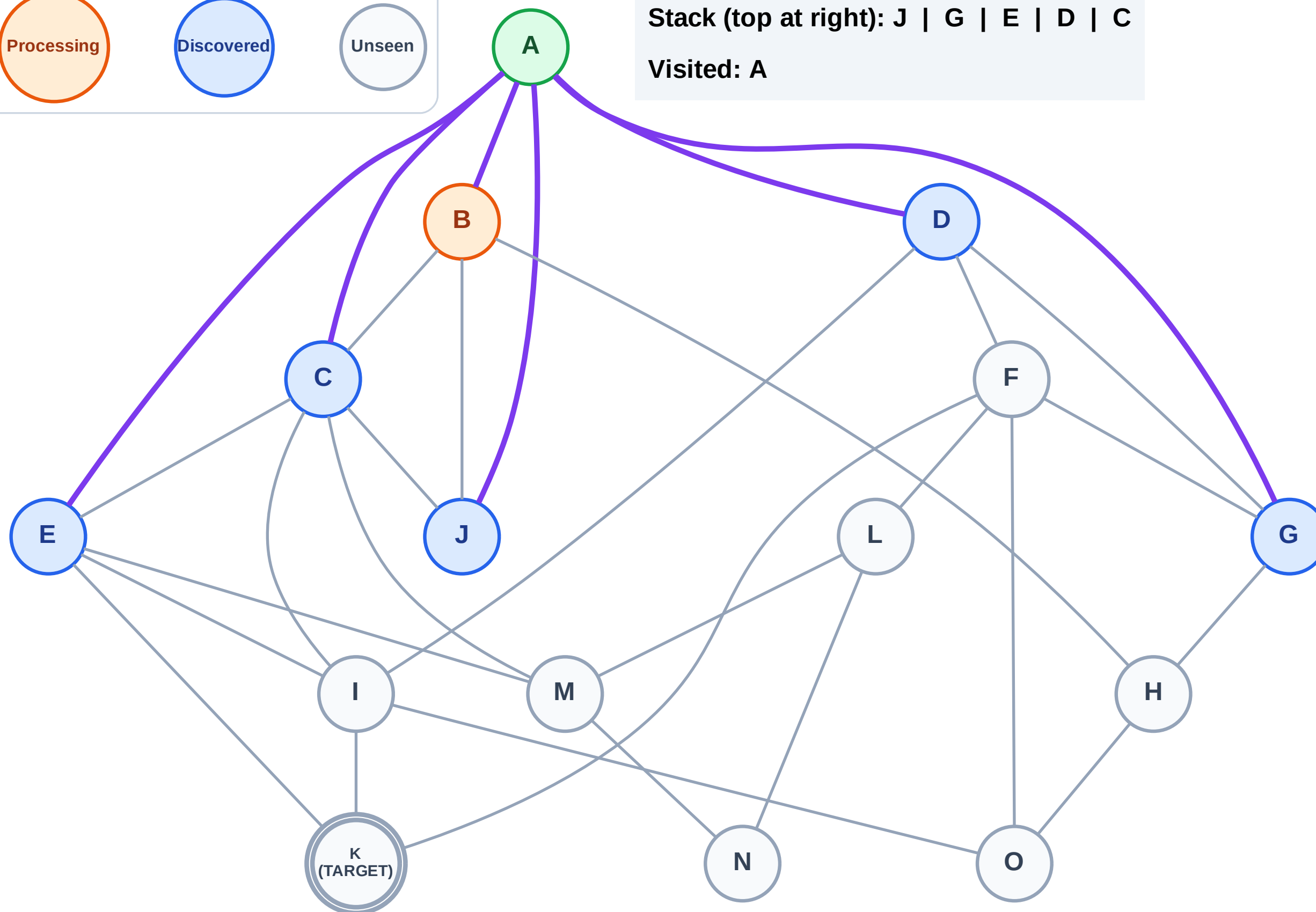


DFS Traversal

Step 4: Process node B



Stack (top at right): J | G | E | D | C
Visited: A



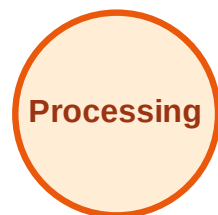
DFS Traversal

Step 5: Discover H from B

Legend



Complete



Processing



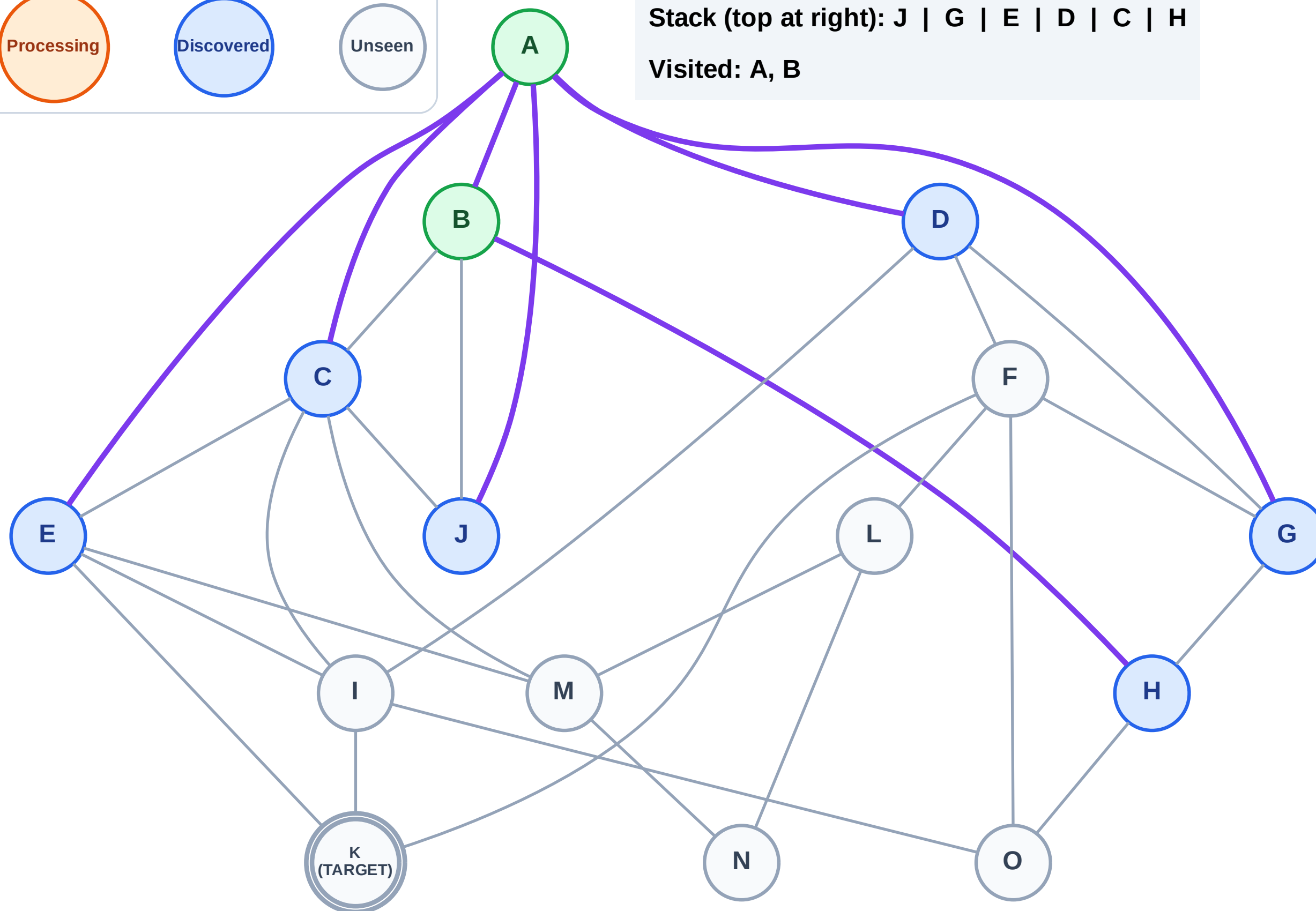
Discovered



Unseen

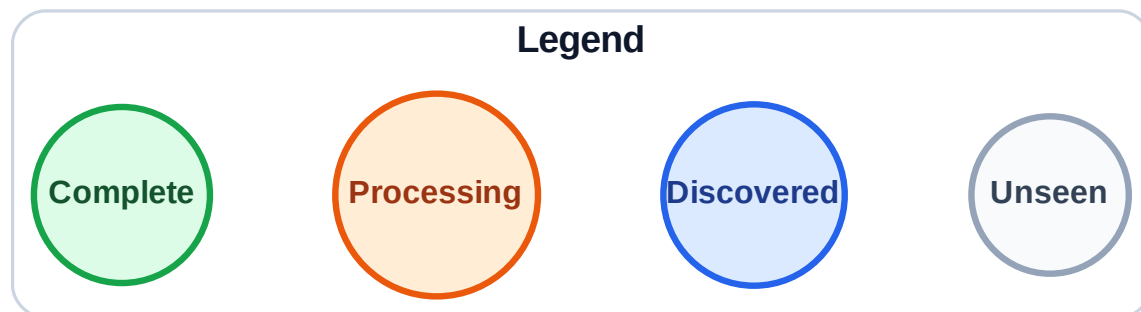
Stack (top at right): J | G | E | D | C | H

Visited: A, B

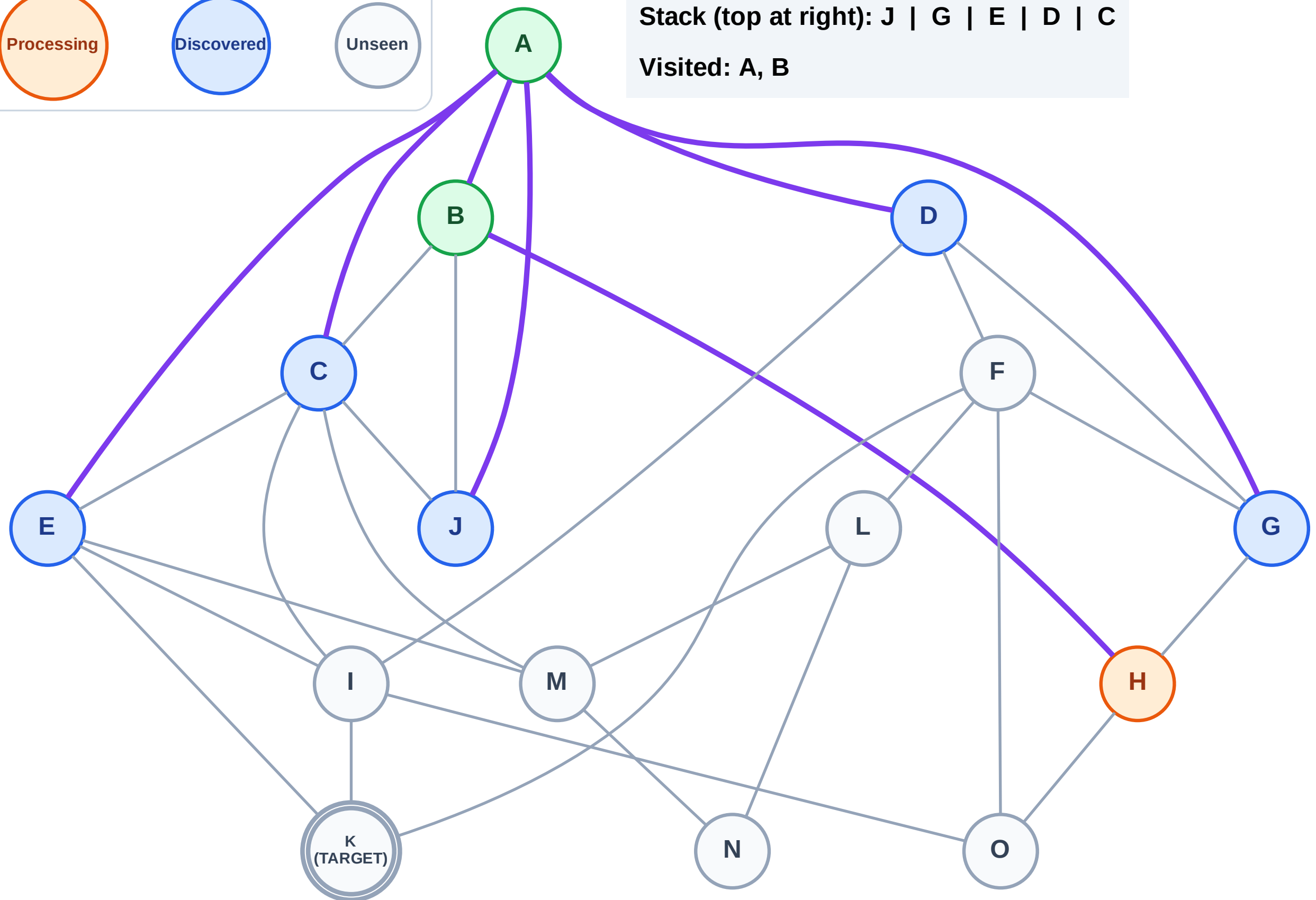


DFS Traversal

Step 6: Process node H



Stack (top at right): J | G | E | D | C
Visited: A, B



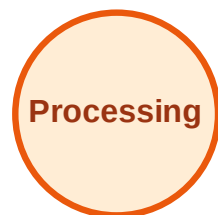
DFS Traversal

Step 7: Discover O from H

Legend



Complete



Processing



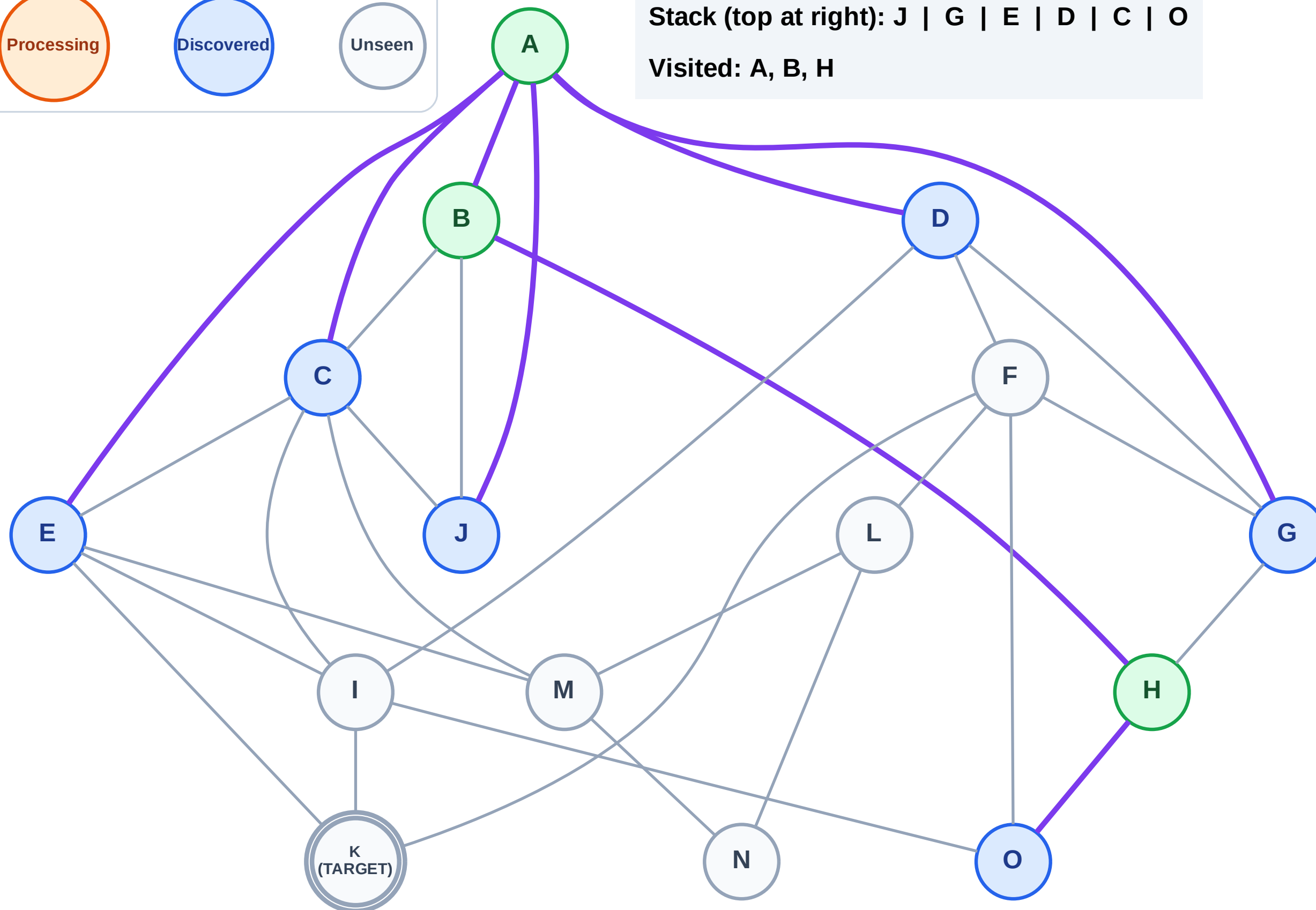
Discovered



Unseen

Stack (top at right): J | G | E | D | C | O

Visited: A, B, H



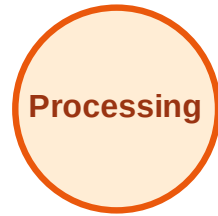
DFS Traversal

Step 8: Process node O

Legend



Complete



Processing



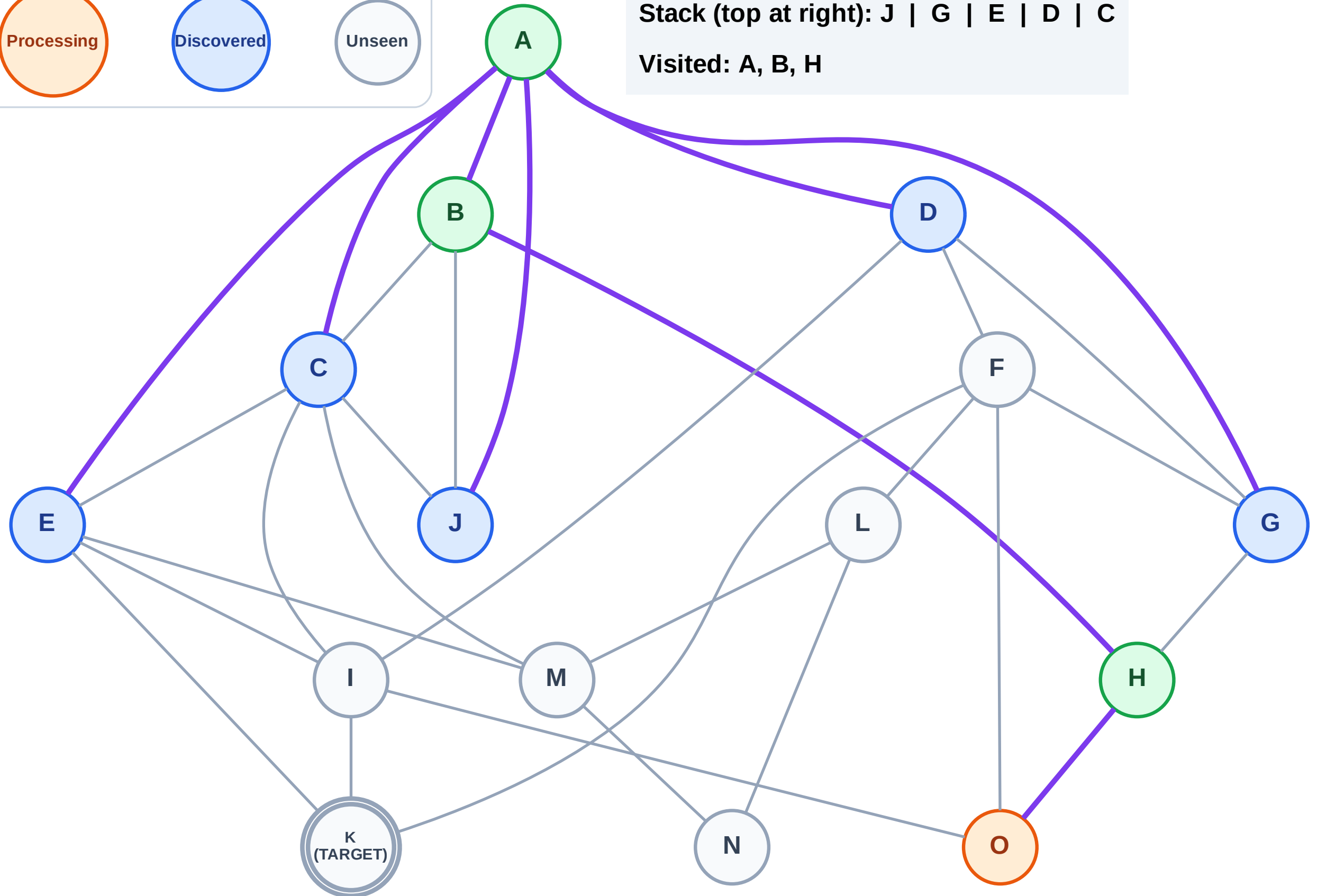
Discovered



Unseen

Stack (top at right): J | G | E | D | C

Visited: A, B, H



DFS Traversal

Step 9: Discover I, F from O

Legend

Complete

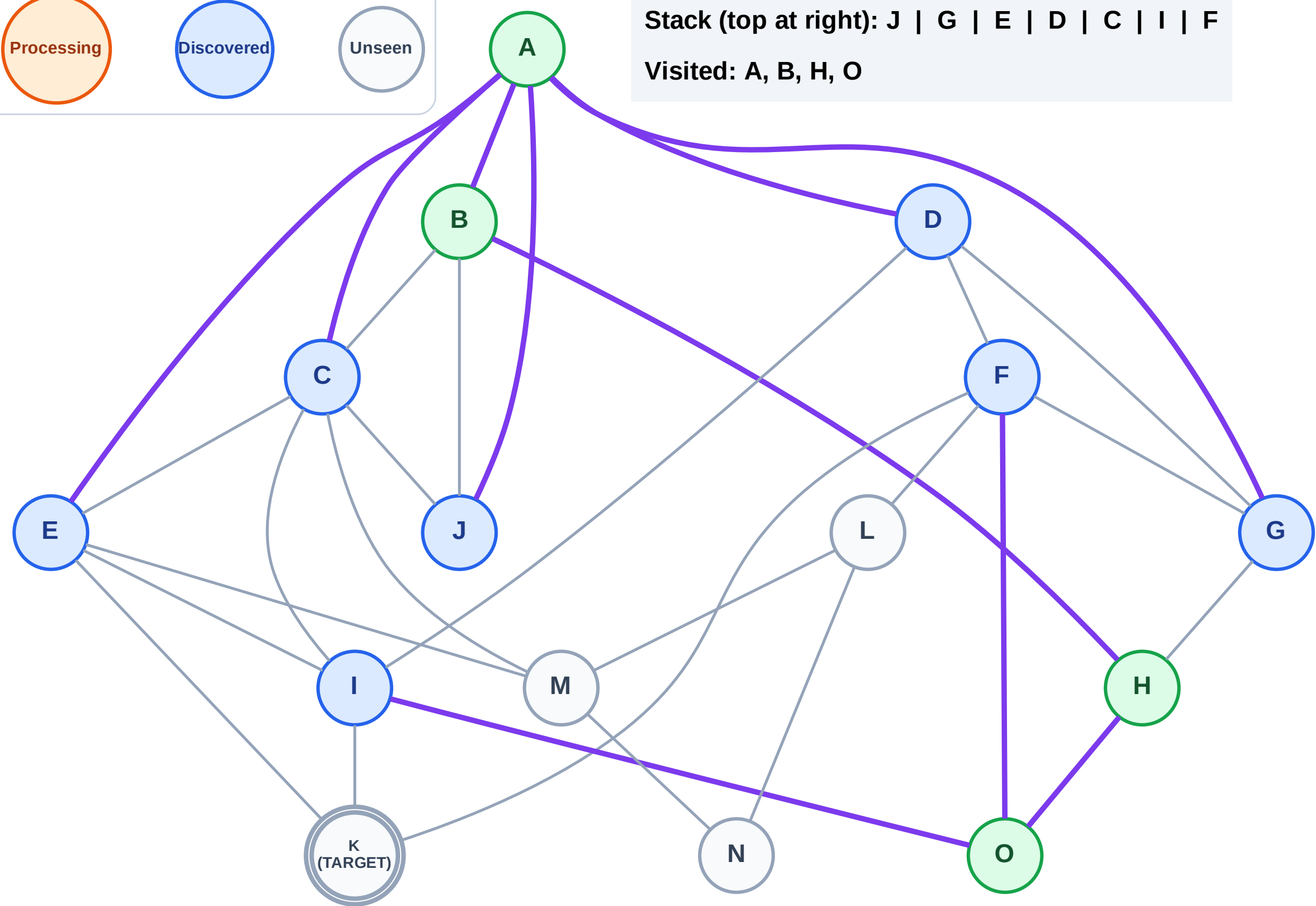
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | I | F

Visited: A, B, H, O



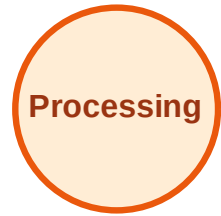
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



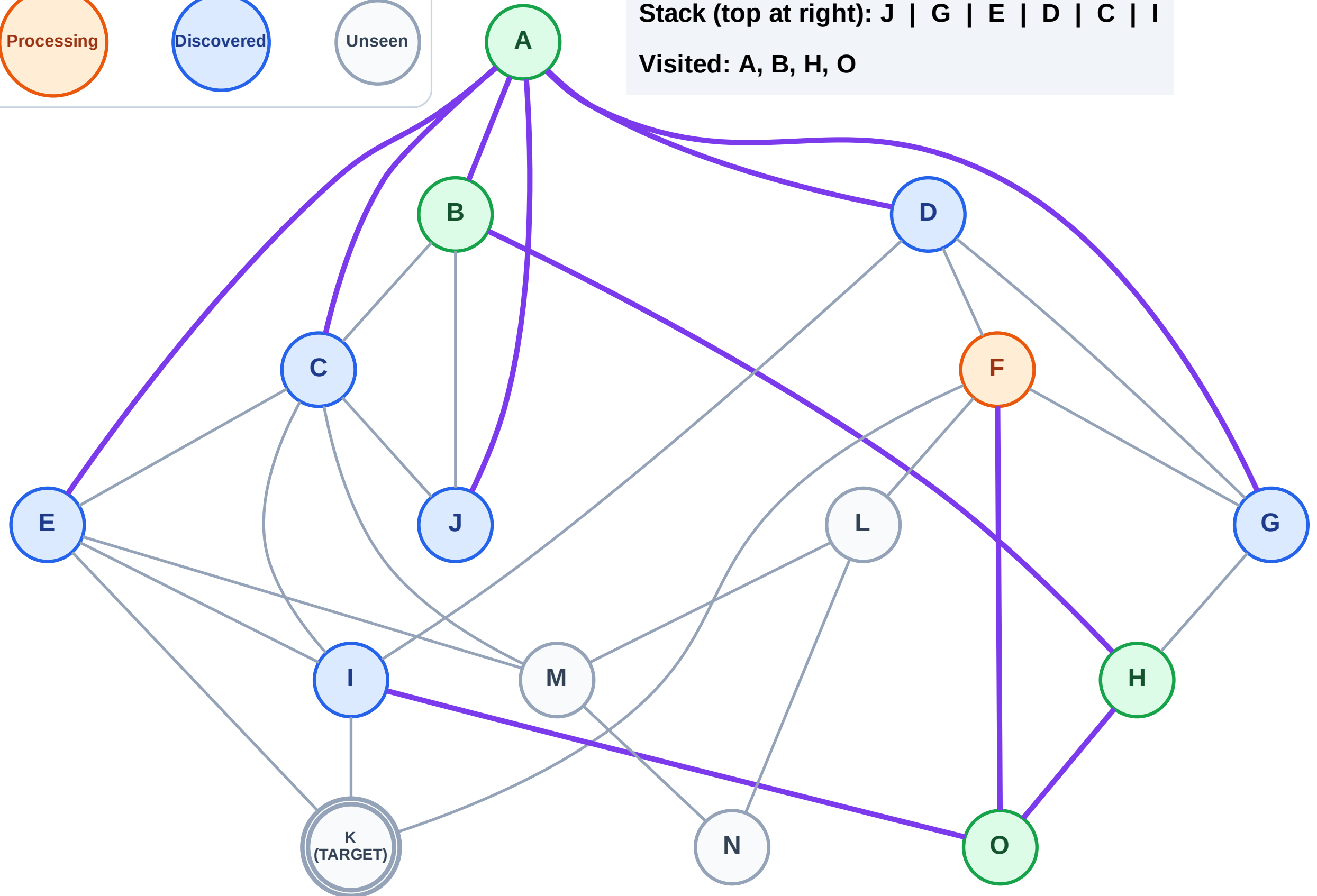
Discovered



Unseen

Stack (top at right): J | G | E | D | C | I

Visited: A, B, H, O



DFS Traversal

Step 11: Discover L, K from F

Legend

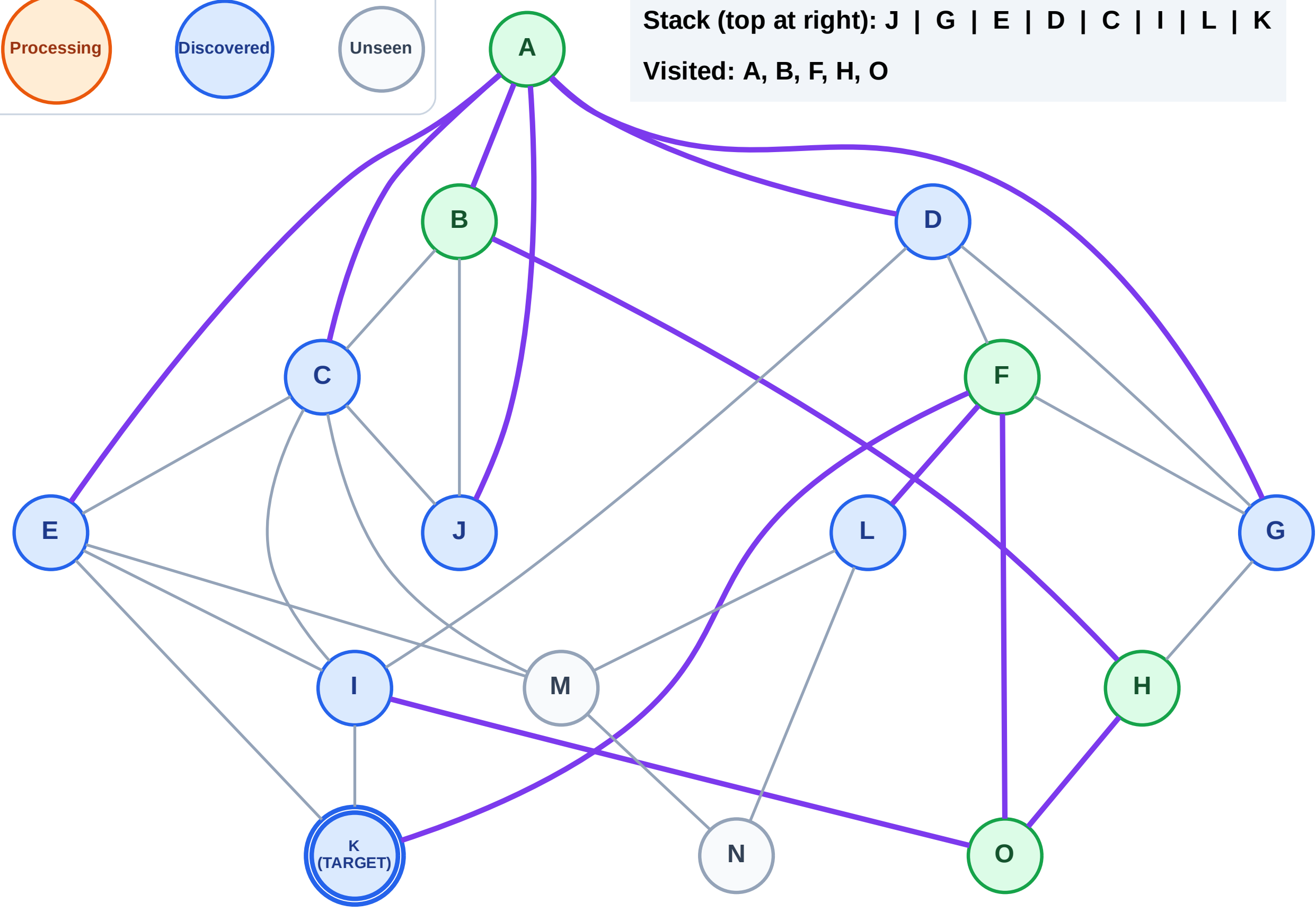
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | I | L | K
Visited: A, B, F, H, O



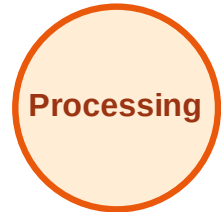
DFS Traversal

Step 12: Process node K

Legend



Complete



Processing



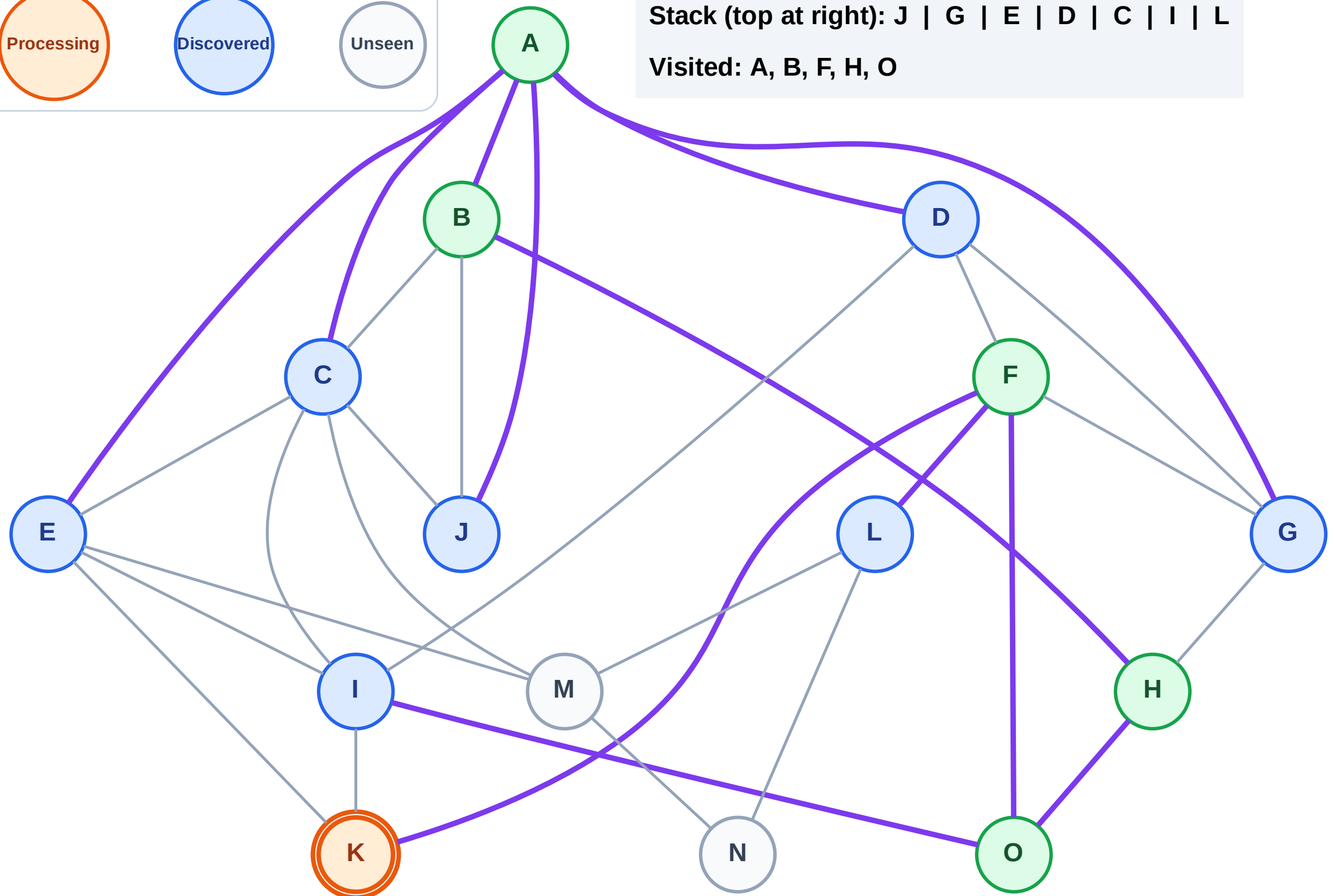
Discovered



Unseen

Stack (top at right): J | G | E | D | C | I | L

Visited: A, B, F, H, O



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → H → O → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | I | L

Visited: A, B, F, H, K, O

